ProntoGel



Sugar

Polymers

Peptides

Biopolymers

Protein



BISCHOFF
CHROMATOGRAPHY

Selection by USP

A rigid, spherical styrenedivinylbenzene copolymer, 3 to 30 µm in diameter.	L21
Packing having the capacity to separate compounds with a molecular weight range from 100-5000 (as determined by polyethylene oxide), applied to neutral, anionic, and cationic water-soluble polymers. A polymethacrylate resin base, cross-linked with polyhydroxylated ether (surface contained some residual carboxyl functional groups) was found suitable.	L25
Packing having the capacity to separate proteins by molecular size over a range of 2,000 to 40,000 Da. it is a polymethacrylate gel.	L37
A methacrylate-base size-exclusion packing for water-soluble samples.	L38
A hydrophilic-polyhydroxymethacrylate gel of totally porous spherical resin.	L39

ProntoGel-AQ-OH

	Particle size (µm)	Type	USP
ProntoGel-AQ-OH 200 Separation Range 100Da-20 000 Da	10	spherical	L25, L38, L39
ProntoGel-AQ-OH 250 Separation Range 100Da-70 000 Da	10	spherical	L37, L38, L39
ProntoGel-AQ-OH 300 Separation Range 1 000Da-300 000Da	10	spherical	L37, L38, L39
ProntoGel-AQ-OH 350 Separation Range 2 500Da-1 000 000Da	10	spherical	L38, L39
ProntoGel-AQ-OH 400 Separation Range 10 000Da-5 000 000Da	10	spherical	L38, L39
ProntoGel-AQ-OH 450 Separation Range 50 000Da →10 000 000Da	10	spherical	L38, L39

ProntoGel-P

	Particle size (µm)	Type	USP
ProntoGel-P-50 Separation Range 100Da-2 500Da	5, (10)	spherical	L21
ProntoGel-P-100 Separation Range 100Da-10 000Da	5, (10)	spherical	L21
ProntoGel-P-500 Separation Range 100Da-30 000Da	5, (10)	spherical	L21
ProntoGel-P-1.000 Separation Range 100Da-70 000Da	5, (10)	spherical	L21
ProntoGel-P-1.500 Separation Range 100Da-120 000Da	5, (10)	spherical	L21
ProntoGel-P-10 000 Separation Range 100Da-400 000Da	5, (10)	spherical	L21
ProntoGe-P-100 000 Separation Range 500Da-1 500 000Da	5, (10)	spherical	L21
ProntoGel-P-1000 000 Separation Range 10 000Da-4 000 000Da	5, (10)	spherical	L21
ProntoGel-P-10 000 000 Separation Range 100 000Da-10 000 000Da	5, (10)	spherical	L21

Selection by Applications

ProntoGel Su			
	Particle size (µm)	Type	For analysis of
ProntoGel Su Ca I	10	spherical	sugars sugar alcohols alcohols
ProntoGel Su Ca II			
ProntoGel Su H I	10	spherical	sugars sugar alcohols alcohols carboxylic acids
ProntoGel Su H II			
ProntoGel Su Pb	10	spherical	sugars especially wood hydrolysates

ProntoGel-Oilgo-Su			
	Particle size (µm)	Type	For analysis of
ProntoGel Oligo Su Na	10	spherical	sugars sugar alcohols alcohols
ProntoGel Oligo Su Ag	10	spherical	sugars sugar alcohols alcohols

ProntoGel-DMF			
	Particle size (µm)	Type	For GPC/SEC analysis of
ProntoGel-100-DMF Separation Range 100Da-2 500Da	10	spherical	PMMA PAN cellulose soluble polymers for use in DMF, DMAc and NMP
ProntoGel-200-DMF Separation Range 100Da-20 000Da	10	spherical	
ProntoGel-250-DMF Separation Range 100Da-70 000Da	10	spherical	
ProntoGel-300-DMF Separation Range 1 000Da-300 000Da	10	spherical	
ProntoGel-350-DMF Separation Range 1 000Da-1 000 000Da	10	spherical	
ProntoGel-400-DMF Separation range 10 000Da-5 000 000Da	10	spherical	
ProntoGel-450-DMF Separation range 100 000Da→10 000 000Da	10	spherical	

ProntoGel AC-AQ			
	Particle size (µm)	Type	For GPC/SEC analysis of
ProntoGel-150-AC-AQ	10	spherical	For GPC analyis in 80/20 aceton/water organosolv lignins calibration: molecular weight vs. PEO/PEG
ProntoGel-250-AC-AQ Separation Range100Da– 70 000Da	10	spherical	
ProntoGel-350-AC-AQ Separation Range100Da–1 000 000Da	10	spherical	

Selection by Applications

ProntoGel-NAC			
	Particle size (µm)	Type	For GPC/SEC analysis of
ProntoGel-100-NAC Separation Range 100Da-2 500Da	10	spherical	Special designed for aqueous GPC/SEC: polycations polyamines (chitosanes) polyethylenoxides Polysaccarides polyanions (heparins, pectins)
ProntoGel-150-NAC Separation Range 100Da - 5 000Da	10	spherical	
ProntoGel-350-NAC Separation Range 2 500Da - 1 000 000Da	10	spherical	
ProntoGel-450-NAC Separation Range 300 000Da - 5 000 000Da	10	spherical	
ProntoGel-500-NAC Separation Range 10 000Da - 50 000 000Da	10	spherical	

ProntoGel-DMSO			
	Particle size (µm)	Type	For GPC/SEC analysis of
ProntoGel-100-DMSO Separation Range 100Da-2 500Da	10	spherical	For analysis in DMSO amylose, amylopectin, starch urea-formaldehyd resins (UF-resins) melamin-urea-formaldehyd resins (MUF-resins) lignins, humic substances, humic acids, coniferous wood bark essences polysaccharide, polysaccharid Derivatives poly(N-isopropylacrylamid) PNIPA Poly-vinylpyridin calibration: pullulan, dextran, polyvinylpyridin et al.
ProntoGel-200-DMSO Separation Range 100Da-20 000Da	10	spherical	
ProntoGel-250-DMSO Separation Range 100Da-70 000Da	10	spherical	
ProntoGel-300-DMSO Separation Range 1 000Da-500 000Da	10	spherical	
ProntoGel-350-DMSO Separation Range 5 000Da-1 500 000Da	10	spherical	
ProntoGel-400-DMSO Separation Range 10 000Da-5 000 000Da	10	spherical	
ProntoGel-450-DMSO Separation Range 50 000Da-10 000 000Da	10	spherical	
ProntoGel-600-DMSO Separation Range →20 000 000Da	10	spherical	
ProntoGel-MP-DMSO Separation Range 100Da-1 000 000Da	10	spherical	

ProntoGel HFIP			
	Particle size (µm)	Type	For GPC/SEC analysis of
ProntoGel-100-HFIP Separation Range 100Da-2 500Da	7, 10	spherical	For GPC analyis in HFIP Polyesters (polybutylene terephthalate / PBTpolyethylene terephthalate /PET, polylactide PLA/ Polyamide 6 / PA6/ polyamide 6-6 PA6-6 / polyhexamethylene adipamide / polyamide 6-10 / PA6-10 / poly(hexamethylene sebacamide) / PA 6-10 Other (paraformaldehyde /polyoxymethylene POM / polyacetal / polyethylenimine / PEI / poly (iminoethylene / polyaziridine calibration: molecular weight vs. PEO
ProntoGel-350-HFIP Separation Range up to 1 000 000 Da	7, 10	spherical	
ProntoGel-500-HFIP Separation Range 10 000Da- ~50 000 000Da	7, 10	spherical	

ProntoGel-DMF

ProntoGel GPC columns for GPC analyzes organic (DMAc, DMF and NMP)

Spherical porous polymeric GPC chromatography phases

- Wide range of pore and particle sizes
- Optimal for 0.5ml / min flow rate with 8mm ID columns (40-80 °C)
- Pressure stability 50-150bar, depending on the porosity
- High resolution due to high pore volume
- Long service life
- High reproducibility
- High purity of the GPC column for good interaction-free GPC
- Please avoid: drying out
- Molar mass range: 100Da-10 000 000Da
- Area of applications: PMMA, PAN, cellulose, DMF and DMAc soluble polymers

ProntoGel DMF molar mass range:

P-100	100Da-2 500Da
P-200	100Da-20 000Da
P-250	100Da-70 000Da
P-250 ER*)	100Da-100 000Da
P-300	1 000Da-300 000Da
P-350	1 500Da-1 000 000Da
P-400	10 000Da-5 000 000Da
P-450	100 000Da-→10 000 000Da

*) ER Pore: extended linear range
To cover a very wide range of molecular sizes, GPC columns of appropriate porosity can be combined.

Description	Separation Range
ProntoGel-100-DMF	100Da-2 500Da
ProntoGel-200-DMF	100Da-20 000Da
ProntoGel-250-DMF ER	100Da-70 000Da
ProntoGel-250-DMF	100Da-100 000Da
ProntoGel-300-DMF	1 000Da-300 000Da
ProntoGel-350-DMF	1 000Da-1 000 000Da
ProntoGel-400-DMF	10 000Da-5 000 000Da
ProntoGel-450-DMF	100 000Da-→10 000 000Da

* Note:- (All above mentioned phases are available in 300x8, 250x8, 50x8, 30x8 dimension)

ProntoGel-DMSO

Special GPC/SEC media for fast, accurate and robust GPC-analysis in DMSO

For GPC / SEC analysis in DMSO, Area of applications:

- amylose, amylopectin, starch
- urea-formaldehyd resins (UF-resins)
- melamin-urea-formaldehyd resins (MUF-resins)
- lignins, humic substances, humic acids, coniferous wood bark essences
- polysaccharide, polysaccharid derivatives
- poly (N-isopropylacrylamid) PNIPA
- poly-vinylpyridin
- calibration: pullulan, dextran, polyvinylpyridin et al.

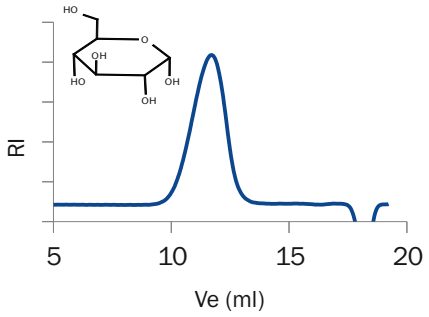
Benefits

- Optimized for DMSO-GPC applications
- Interaction free pure GPC/SEC
- Easy, reliable and robust GPC/SEC-calibration by dextrans, pullulans et al.
- Low column bleeding for low detectornoise for improved light scattering or viscosity detection
- 12µ particle technology for low back pressure
- Large pore volume and optimized mass transfer for polymers giving optimized resolution
- Low costs caused by long live time of column – result of combination of optimized proprietary particle and packing technology.

Description	Separation Range
ProntoGel-100-DMSO	100Da-2 500Da
ProntoGel-200-DMSO	100Da-20 000Da
ProntoGel-250-DMSO	100Da-70 000Da
ProntoGel-300-DMSO	1.000Da-300 000Da
ProntoGel-350-DMSO	1 000Da-1 000 000Da
ProntoGel-400-DMSO	10 000Da-5 000 000Da
ProntoGel-450-DMSO	100 00Da-→10 000 000Da
ProntoGel-600-DMSO	Up to 20 000 000Da
ProntoGel-MP-DMSO	100Da-1 000 000Da

* Note:- (All above mentioned phases are available in 300x8, 250x8, 50x8, 30x8 dimension)

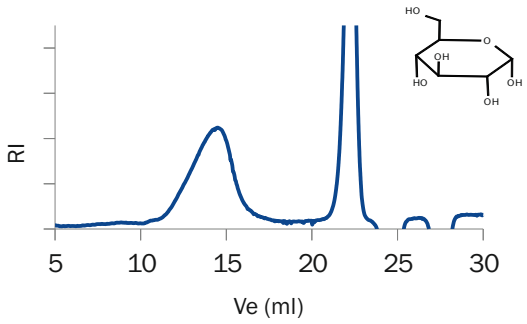
ProntoGel-DMSO



Analyte: **Polysaccharide**
(M ca. 70 000Da)

Column: ProntoGel-250-DMSO
ProntoGel-350-DMSO

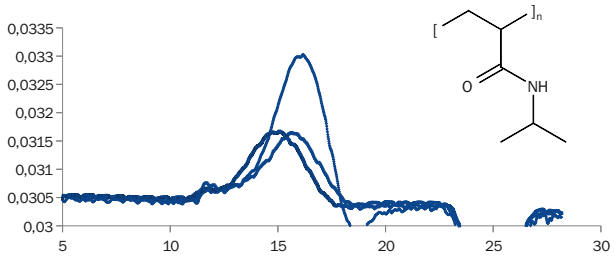
Dimension: ea. 300mm x 8mm
Mobil Phase: DMSO / 0.075M $NaNO_3$
Flow: 0.5ml/min
Temperature: 80°C
Detection: RI



Analyte: **Dextran 650**
Dextran from Leuconostoc spp.,
M = 450 000-650 000Da + fructose

Column: ProntoGel-200-DMSO
ProntoGel-250-DMSO
ProntoGel-350-DMSO

Dimension: ea. 300mm x 8mm
Mobil Phase: DMSO / 0.075M $NaNO_3$
Flow: 0.5ml/min
Temperature: 80°C
Detection: RI

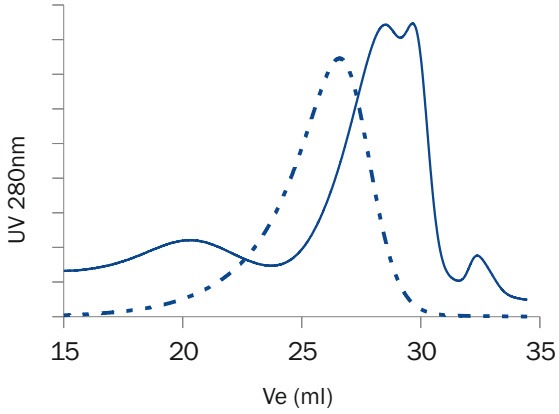


Analyte: **Poly(N-isopropylacrylamide)**
Further denominations: PNIPA, PNIPAAm, NIPA, PNIPAA or PNIPAm.
CAS [25189-55-3], formula: $(C_6H_{11}NO)_n$

3 different PNIPA fractions

Column: ProntoGel-300-DMSO

Dimension: ea. 300mm x 8mm
Mobil Phase: DMSO / 0.075M $NaNO_3$
Flow: 0.5ml/min
Temperature: 80°C
Detection: RI

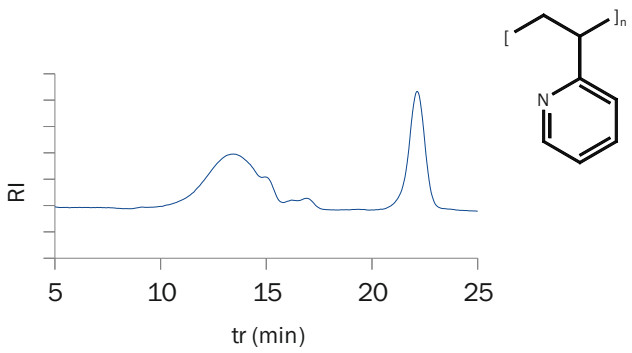


Analyte: **Humic acids / humates GPC**
(range: 100-1 500 000Da)
SEC / GPC comparison GPC / SEC comparison of 2 different humic acids / humates

Column: ProntoGel DMSO-Multipore

Dimension: 3 x 300mm x 8mm
Mobile Phase: DMSO
Flow: 0.4ml/min
Temperature: 70°C
Detection: UV 280nm

ProntoGel-DMSO

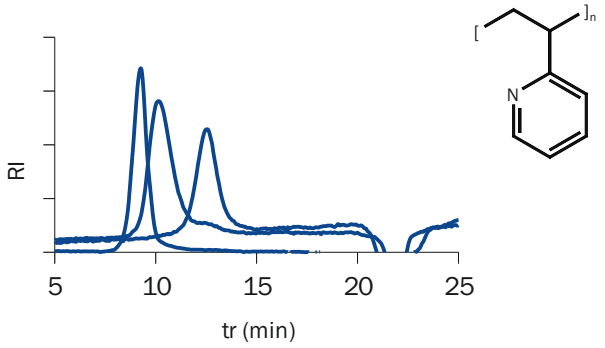


Analyte: **Polyvinylpyridine degradated**

Further denominations: CAS 25014-15-7, $(C_7H_7N)_n$

low molecular weight (oligomeric) polyvinylpyridin fraction

Column:	ProntoGel-250-DMSO
Dimension:	ea. 300mm x 8mm
Mobil Phase:	DMSO / 0.075M $NaNO_3$
Flow:	0.4ml/min
Temperature:	50°C
Detection:	RI



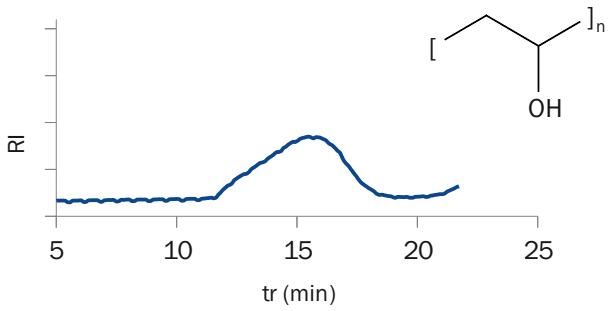
Analyte: **Polyvinylpyridine fractions**

Further denominations: CAS 25014-15-7, $(C_7H_7N)_n$

75.7kDa, 20.9kDa, 3.2kDa

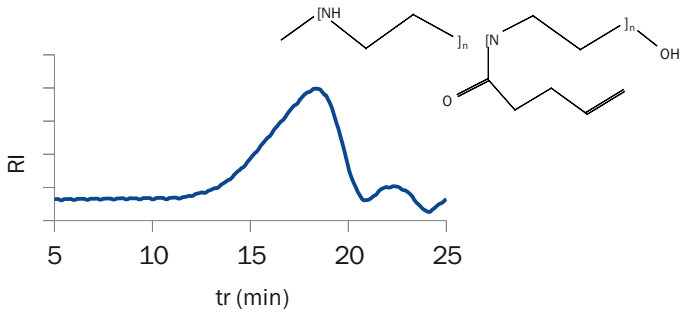
Superposition of 3 different polyvinylpyridin fractions

Column:	ProntoGel-250-DMSO
Dimension:	ea. 300mm x 8mm
Mobil Phase:	DMSO / 0.075M $NaNO_3$
Flow:	0.4ml/min
Temperature:	50°C
Detection:	RI



Analyte: **Polyvinylalcohol M=22kDa**

Column:	ProntoGel-300-DMSO
Dimension:	2 x 300mm x 8mm
Mobil Phase:	DMSO / 0.075M $NaNO_3$
Flow:	0.4ml/min
Temperature:	50°C
Detection:	RI

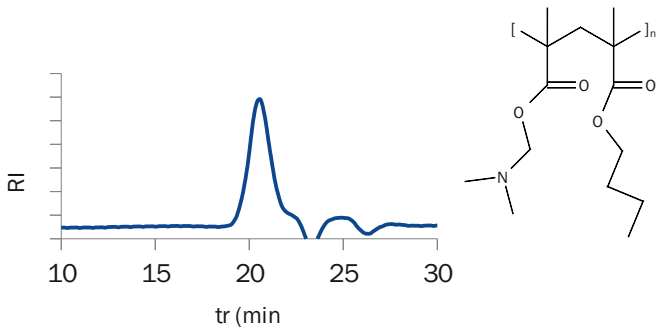


Analyte: **Poly[2-(butenyl)2-oxazoline-co-ethylenimine]**

M = 50.000Da

Column:	ProntoGel-300-DMSO
Dimension:	ea. 300mm x 8mm
Mobil Phase:	DMSO / 0.075M $NaNO_3$
Flow:	0.4ml/min
Temperature:	50°C
Detection:	RI

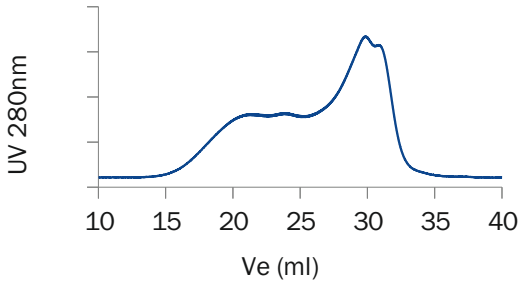
ProntoGel-DMSO



Analyte: **Polybutyl methacrylate/ Poly(dimethylamino-ethylmethacrylate) M=22kDa**

Column: ProntoGel-300-DMSO

Dimension: 2 x 300mm x 8mm
Mobil Phase: DMSO / 0.075M $NaNO_3$
Flow: 0.4ml/min
Temperature: 50°C
Detection: RI

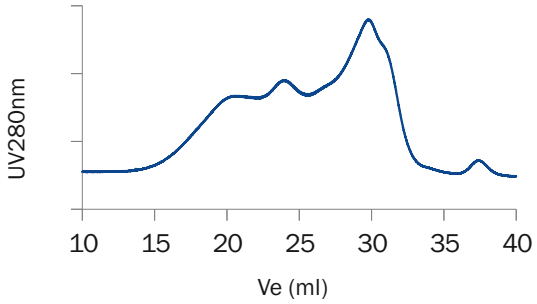


Analyte: **Pea protein GPC**

covering the calibration range of 100-1 000 000Da (based on dextrans)

Column: ProntoGel-MP-DMSO

Dimension: 3 x 300mm x 8mm
Mobil Phase: DMSO
Flow: 0.4ml/min
Temperature: 55°C
Detection: UV 280nm
Calibration: vs. Dextran, pullulan or protein

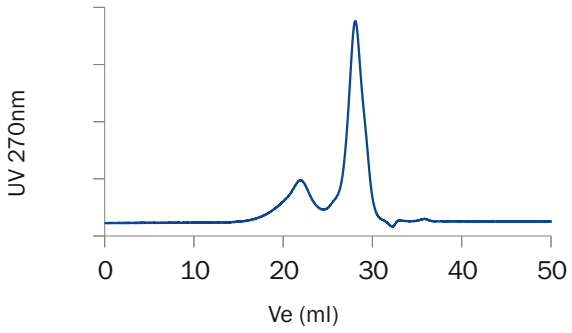


Analyte: **Soy protein GPC**

covering the calibration range of 100-1 000 000Da (based on dextrans)

Column: ProntoGel-MP-DMSO

Dimension: 3 x 300mm x 8mm
Mobil Phase: DMSO
Flow: 0.4ml/min
Temperature: 55°C
Detection: UV 280nm
Calibration: vs. Dextran, pullulan or protein



Analyte: **Manuka honey protein GPC**

covering the calibration range of 100-1 000 000Da (based on dextrans)

Column: ProntoGel-MP-DMSO

Dimension: 3 x 300mm x 8mm
Mobil Phase: DMSO
Flow: 0.4ml/min
Temperature: 40°C
Detection: UV 270nm
Calibration: vs. Dextran, pullulan or protein

ProntoGel-AQ-OH

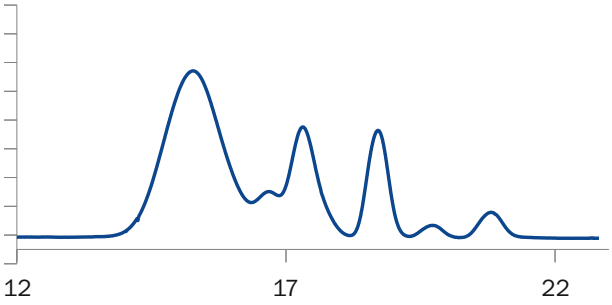
- Optimized for aqueous GPC/SEC-separations
- Aqueous high hydrophilic polymeric base material
- Low column bleeding for low detector noise
- 7µ particle technology (standard) for high platecounts and high resolution*)
- Large pore volume for high resolution
- pH stable 2.5-12
- Pressure stability 30-80 bar (depending on pore size)
- Temperature stability 10-85 °C
- Individual pore sizes for individual molecular weight ranges
- Multi-pore technology for broad range of molecular size
- Increased live of GPC/SEC columns by combination of proprietary particle and packing technology service-application center for method screening available in Bischoff centre.

*) 7µ Particle technology is standard for the small porous series – 100 and – 200.

Description	Separation Range
ProntoGel-AQ-OH 100	100Da-2 500Da
ProntoGel-AQ-OH 200	100Da-20 000Da
ProntoGel-AQ-OH 250	100Da-70 000Da
ProntoGel-AQ-OH 300	1 000Da-300 000Da
ProntoGel-AQ-OH 350	2 500Da-1 000 000Da
ProntoGel-AQ-OH 400	10 000Da-5 000 000Da
ProntoGel-AQ-OH 450	50 000Da→10 000 000Da
ProntoGel-AQ-OH MP	100Da-1 000 000Da

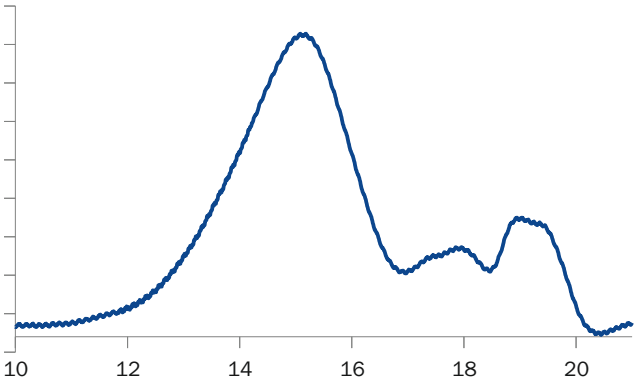
* Note:- (All above mentioned phases are available in 300x8, 250x8, 50x8, 30x8 dimension)

ProntoGel-AQ-OH



Analyte: **Oligosaccharide**

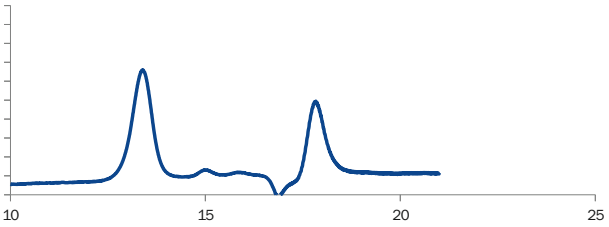
Column:	ProntoGel-AQ-OH 100
Dimension:	2 x 300mm x 8mm
Mobil Phase:	H_2O
Flow:	0.5ml/min
Temperature:	40°C
Detection:	RI
Injection:	20µl sample



Analyte: **Pectin**

sample with high content of oligomers

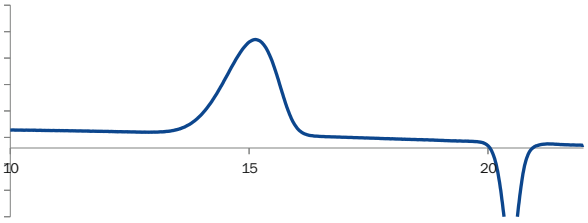
Column:	ProntoGel-AQ-OH 100 ProntoGel-AQ-OH 350
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	H_2O, $NaKHPO_4$ (pH 6.8 0.07M) + 50mM NaCl
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample



Analyte: **PEGylated protein**

Separation of an approx. 5 000Da product of a PEGylated protein of approx. 100 000Da

Column:	ProntoGel-AQ-OH 100 ProntoGel-AQ-OH 350
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	0.05% NaN_3 in H_2O
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample

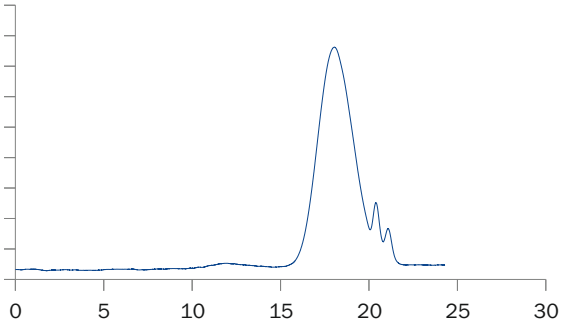


Analyte: **Starch hydrolysate**

analysis of a 100 000Da fraction

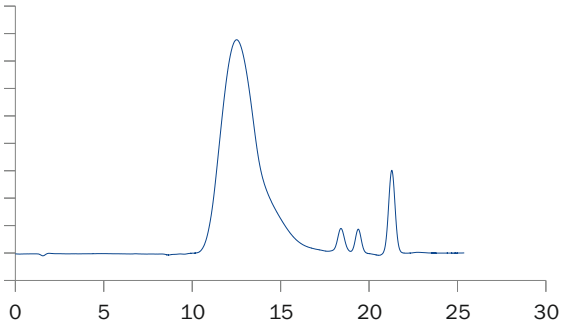
Column:	ProntoGel-AQ-OH 100 ProntoGel-AQ-OH 350
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	0.2M $NaNO_3$ in H_2O
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample

ProntoGel-AQ-OH



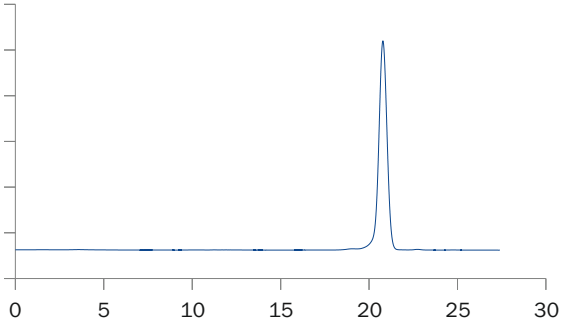
Analyte: **Inulin**
analysis of a 100Da- 1 000 000Da including oligomer separation

Column:	ProntoGel-AQ-OH 250 ProntoGel-AQ-OH 350
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	0.075M $NaNO_3$, 5g/l $Na_2HPO_4x7H_2O$ in H_2O
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample



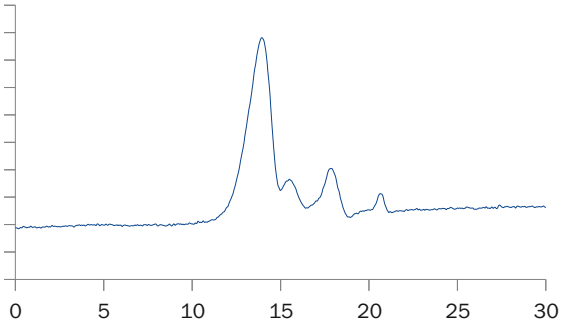
Analyte: **Hyaluronic acid**
including oligomer separation

Column:	ProntoGel-AQ-OH 250 ProntoGel-AQ-OH 350
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	0.075M $NaNO_3$, 5g/l $Na_2HPO_4x7H_2O$ in H_2O
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample



Analyte: **Corn syrup**
analysis of a 100Da- 1 000 000Da area

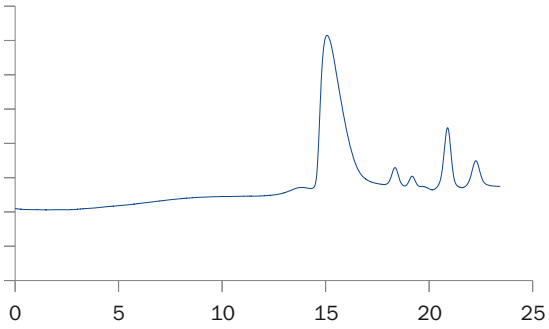
Column:	ProntoGel-AQ-OH 250 ProntoGel-AQ-OH 350
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	H_2O
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample



Analyte: **Water-based acrylate dispersion**
analysis of a 100Da- 1 000 000Da area

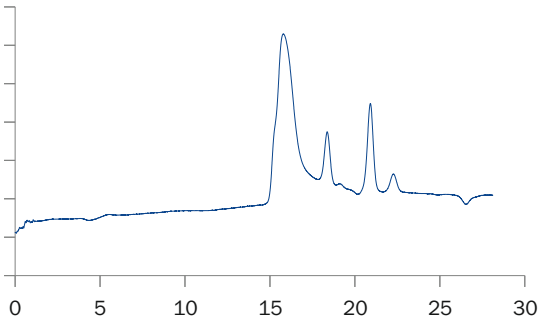
Column:	ProntoGel-AQ-OH 250 ProntoGel-AQ-OH 350
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	1.0M $NaNO_3$ in H_2O
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample

ProntoGel-AQ-OH



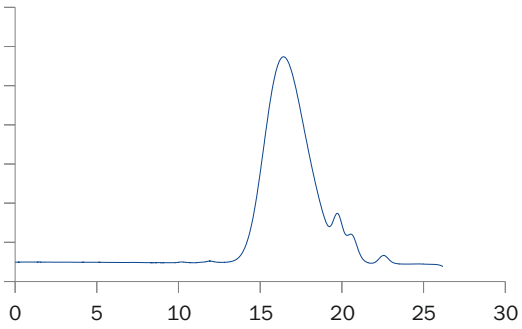
Analyte: **Heparin-Na, 8-25kDa**
including oligomer separation

Column:	ProntoGel-AQ-OH 250 ProntoGel-AQ-OH 350
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	0.075M $NaNO_3$, 5g/l $Na_2HPO_4x7H_2O$ in H_2O
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample



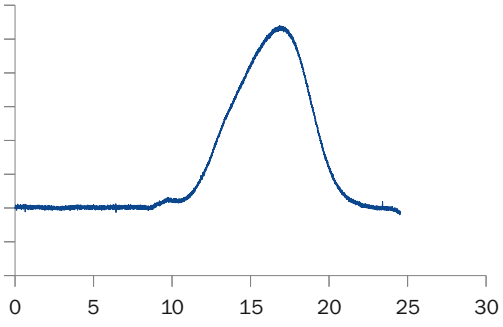
Analyte: **Dextran sulfate-Na**
analysis of a 100Da- 1 000 000Da area

Column:	ProntoGel-AQ-OH 250 ProntoGel-AQ-OH 350
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	0.075M $NaNO_3$, 5g/l $Na_2HPO_4x7H_2O$ in H_2O
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample



Analyte: **Alginate-Na**
analysis of a 100Da- 1 000 000 Da area

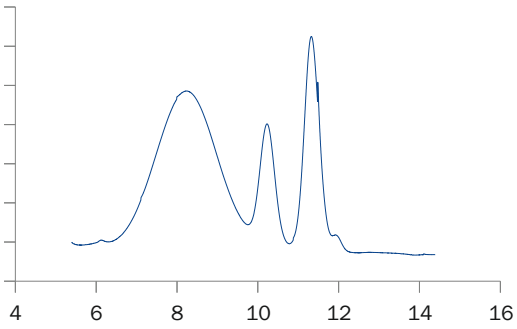
Column:	ProntoGel-AQ-OH 250 ProntoGel-AQ-OH 350
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	0.075M $NaNO_3$, 5g/l $Na_2HPO_4x7H_2O$ in H_2O
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample



Analyte: **Carrageenan**
analysis of a 1 000Da- 5 000 000Da area

Column:	ProntoGel-AQ-OH 350 ProntoGel-AQ-OH 450
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	0.075M $LiNO_3$ in H_2O
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample

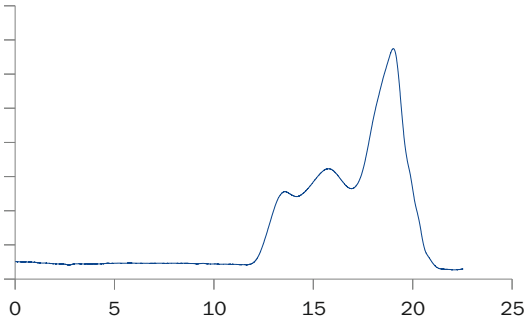
ProntoGel-AQ-OH



Analyte: **Pullulan**
including oligomer separation

Column: ProntoGel-AQ-OH Screening

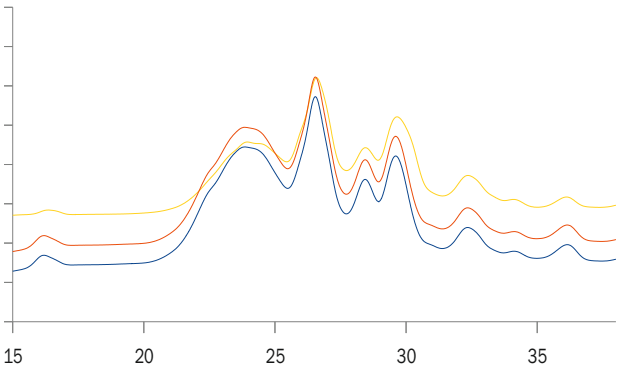
Dimension: 300mm x 8mm
Mobil Phase: 0.075M $NaNO_3$ in H_2O
Flow: 1.0ml/min
Temperature: 20°C
Detection: RI
Injection: 20µl sample



Analyte: **Maltodextrin 12**
analysis of a 100Da-1 000 000Da area

Column: ProntoGel-AQ-OH 250
ProntoGel-AQ-OH 350

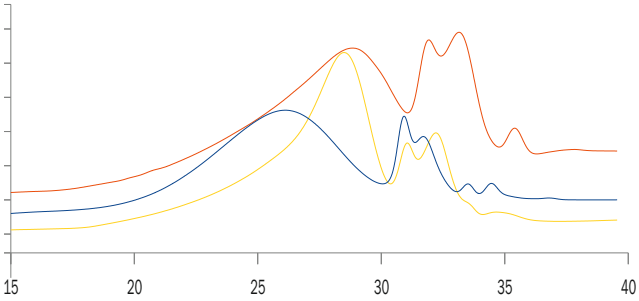
Dimension: e.a. 300mm x 8mm
Mobil Phase: 0.075M $NaNO_3$ in H_2O
Flow: 1.0ml/min
Temperature: 20°C
Detection: RI
Injection: 20µl sample



Analyte: **Fish protein hydrolysates**
3 different batches
analysis of a 100Da-70 000 Da area
including oligomer separation

Column: ProntoGel-AQ-OH 250

Dimension: 3 x 300mm x 8mm
Mobil Phase: 0.05M $NaNO_3$ + 0,07M $Na_{1,5}H_{1,5}PO_4$ in H_2O
Flow: 1.0ml/min
Temperature: 30°C
Detection: RI
Injection: 20µl sample

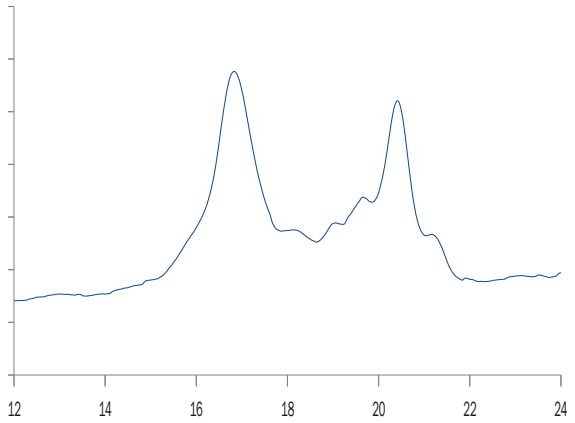


Analyte: **Polycarboxylate ether**
3 different batches

Column: ProntoGel-AQ-OH 350

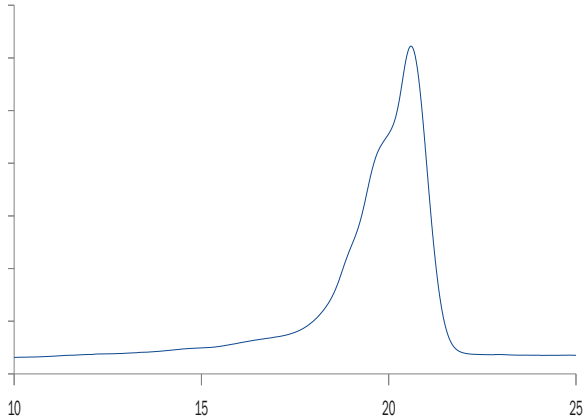
Dimension: 3 x 300mm x 8mm
Mobil Phase: 0.1M $NaNO_3$+0.05M Na_2HPO_4 in H_2O
Flow: 1.0ml/min
Temperature: 20°C
Detection: RI
Injection: 20µl sample

ProntoGel-AQ-OH



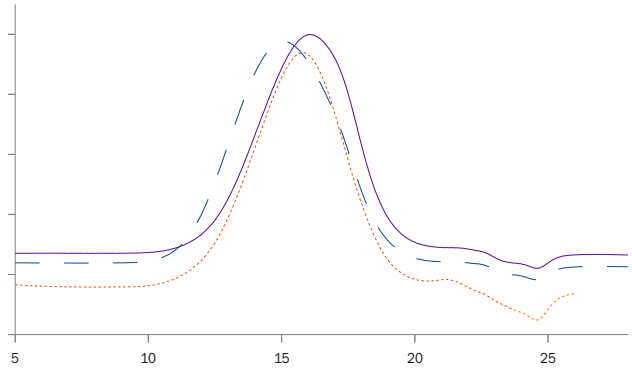
Analyte: **Water-soluble casein fraction**

Column:	ProntoGel-AQ-OH 250 ProntoGel-AQ-OH 350
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	0.01M $Na_{1,5}H_{1,5}PO_4$ & 0.3M NaCl in H_2O
Flow:	1.0ml/min
Temperature:	30°C
Detection:	RI
Injection:	50µl sample



Analyte: **Jelly Bean "Gummibärchen"**

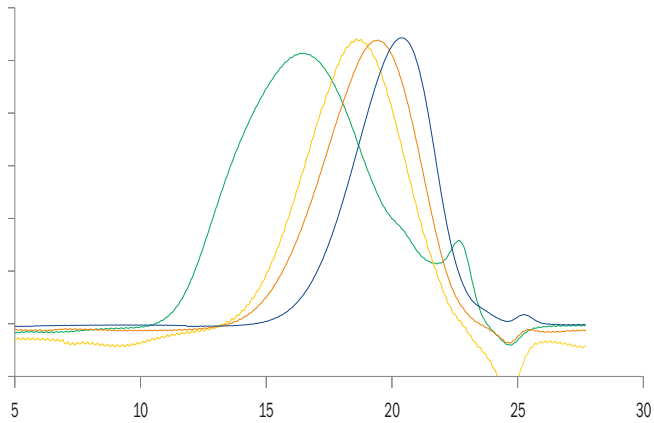
Column:	ProntoGel-AQ-OH 250 ProntoGel-AQ-OH 350
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	0.01M $Na_{1,5}H_{1,5}PO_4$ & 0.3M NaCl in H_2O
Flow:	1.0ml/min
Temperature:	50°C
Detection:	RI
Injection:	20µl sample Ingredient according to label: 6.9% protein (gelatin) dissolved in the eluent



Analyte: **Polyvinyl alcohols, 88% degree of hydrolysis**

3 different batches
including oligomer separation

Column:	ProntoGel-AQ-OH 350
Dimension:	2 x 300mm x 8mm
Mobil Phase:	0.05M Na_2HPO_4 + 0.1M $NaNO_3$ in H_2O
Flow:	1.0ml/min
Temperature:	30°C
Detection:	RI
Injection:	20µl sample

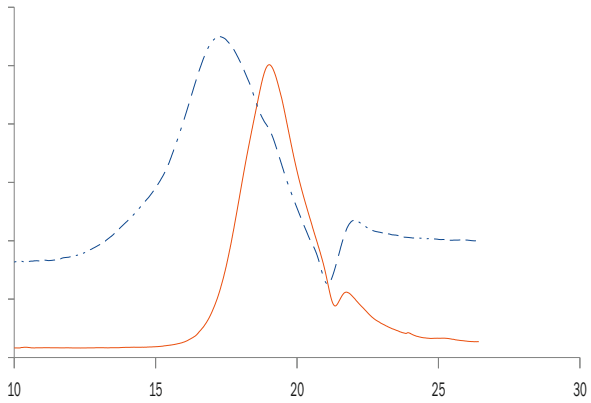


Analyte: **PVP, polyvinyl pyrolidone**

3 different batches
including oligomer separation

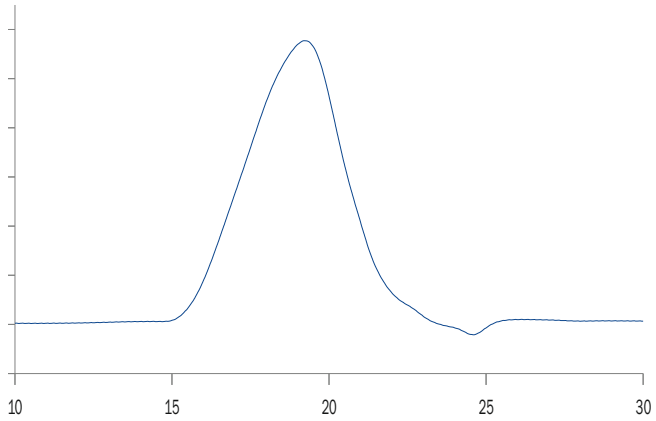
Column:	ProntoGel-AQ-OH 350
Dimension:	2 x 300mm x 8mm
Mobil Phase:	0.1M $NaNO_3$ in H_2O + 20% ACN
Flow:	1.0ml/min
Temperature:	30°C
Detection:	RI
Injection:	50µl sample PVP (M = 8,3 / 23,4 / 33,7 / 175kDa) dissolved in the eluent

ProntoGel-AQ-OH



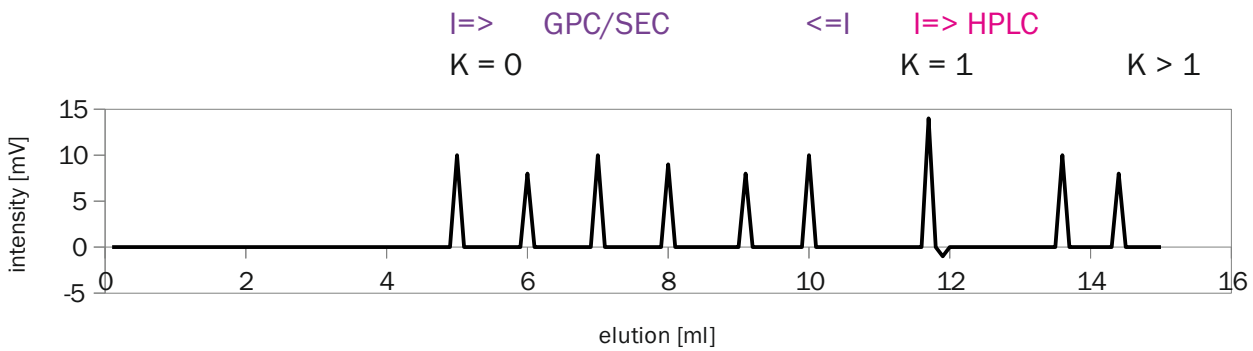
Analyte: **Pork gelatin vs. gelatin from collagen hydrolysate**
including oligomer separation

Column:	ProntoGel-AQ-OH 250 ProntoGel-AQ-OH 350
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	0.01M $Na_{1,5}H_{1,5}PO_4$ & 0.3M NaCl in H_2O
Flow:	1.0ml/min
Temperature:	30°C
Detection:	RI
Injection:	ea. 50µl sample Pork gelatine (blue - - - -), Gelatin from collagen hydrolyzate (red - - - -)



Analyte: **Pork gelatin**
analysis of a 100Da- 1 000 000Da,

Column:	ProntoGel-AQ-OH 250 ProntoGel-AQ-OH 350
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	0.075M $NaNO_3$, 5g/l $Na_2HPO_4x7H_2O$ in H_2O
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample

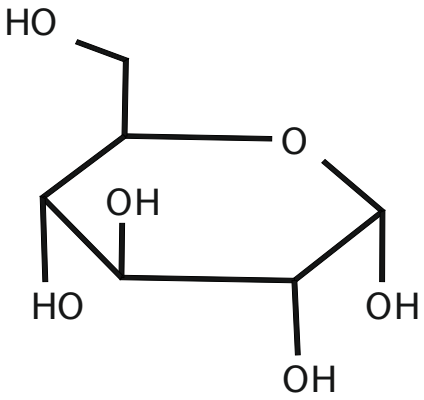


GPC/SEC – Separation by size (ΔS) II HPLC – Separation according to bond strength (ΔH)
K = Partition coefficient

GPC/SEC finishes when HPLC begins

For HPLC-Analyses of

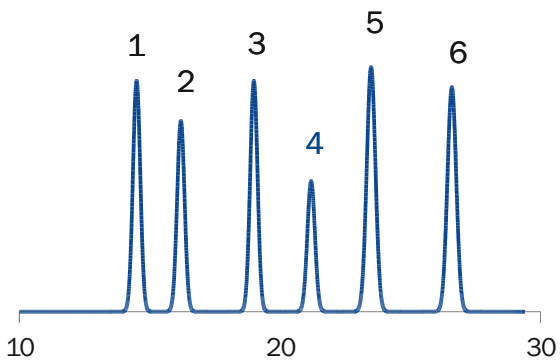
- Sugars/Carbohydrates
- Sugar Alcohols
- Alcohols
- Carboxylic Acids



Advantages:

ProntoGel Su Ca I/II	–	Analysis of sugars, sugar alcohols, alcohols.
ProntoGel Su Pb	–	Analysis of sugars esp. wood extracts
ProntoGel Su H I/II	–	Analysis of sugars, sugaralcohols, alcohols and carobxylic acids.
ProntoGel Su Na	–	Analysis of sugars, sugaralcohols and alcohols.
ProntoGel Oligo Su Na	–	Analysis of sugars, sugaralcohols and alcohols.
ProntoGel Oligo Su Ag	–	Analysis of sugars, sugaralcohols and alcohols.

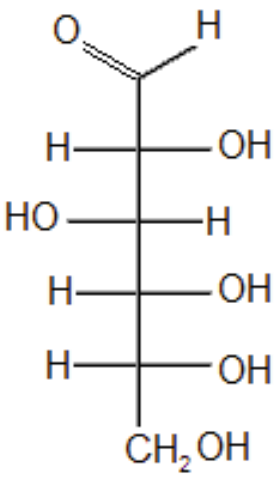
Special Polymer for fast, easy and reliable determination using HPLC-RI or HPLC-ELSD at 60-80°C.



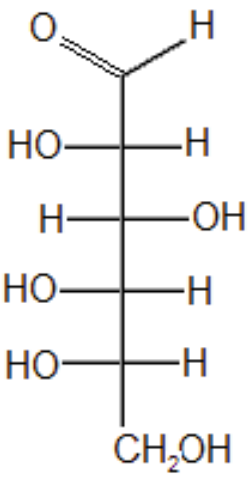
Analyte:	**Mixture of**
	1. Sucrose (Saccarose)
	2. Glucose
	3. Fructose
	4. Glycerin
	5. Mannitol
	6. Sorbitol
Column:	ProntoGel Su ca
Dimension:	300mm x 8mm
Mobil Phase:	H_2O
Flow:	0.5ml/min
Temperature:	80°C
Detection:	RI
Injection:	20µl sample

Identification of many sugars

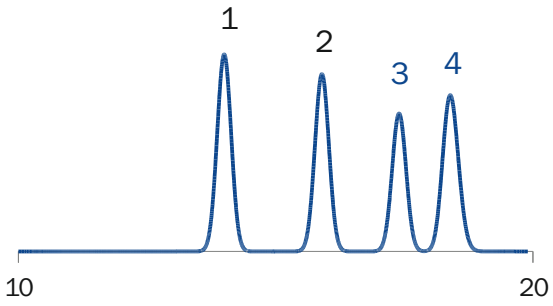
Description	Dimesnsion
ProntoGel Su Ca I	300mm x 8mm 250mm x 8mm 30mm x 8mm
ProntoGel Su Pb	300mm x 8mm 250mm x 8mm 30mm x 8mm
ProntoGel-Su H I (SO_3H)	300mm x 8mm 250mm x 8mm 30mm x 8mm
ProntoGel Su H II (SO_3H)	300mm x 8mm 250mm x 8mm 30mm x 8mm
ProntoGel Su Na	300mm x 8mm 250mm x 8mm 30mm x 8mm



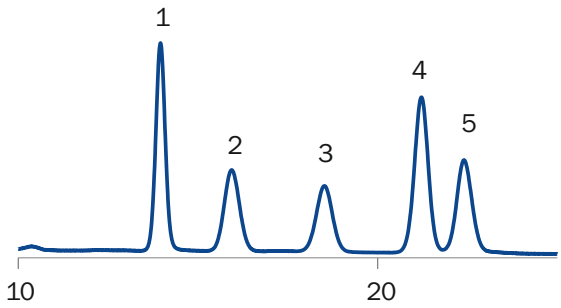
D-Glucose



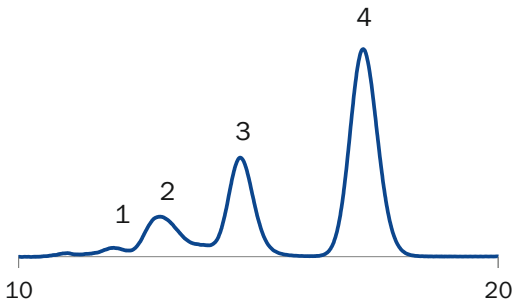
L-Glucose



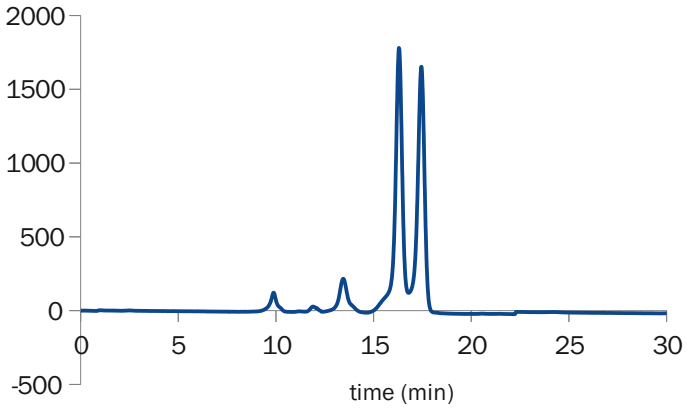
Analyte:	Mixture of
	1. Sucrose (Saccarose)
	2. Glucose
	3. Fructose
	4. Glycerin
Column:	ProntoGel Su Pb
Dimension:	300mm x 8mm
Mobil Phase:	H_2O
Flow:	0.4ml/min
Temperature:	60°C
Detection:	RI
Injection:	20µl sample



Analyte:	Mixture of
	1. Sucrose (Saccarose)
	2. Glucose
	3. Fructose
	4. Glycerin
	5. Ethanol
Column:	ProntoGel Su Ca
Dimension:	300mm x 8mm
Mobil Phase:	H_2O
Flow:	0.5ml/min
Temperature:	80°C
Detection:	RI
Injection:	20µl sample

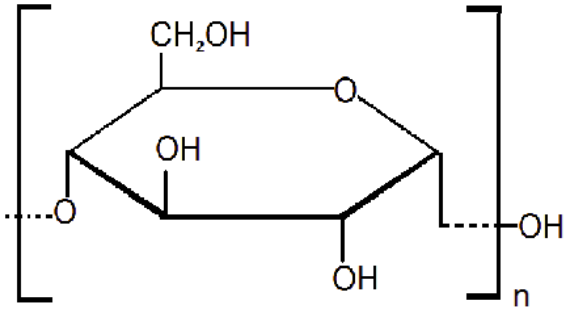


Analyte:	Honey (solvet in H_2O)
	1. Dp 3
	2. Dp 2
	3. Glucose
	4. Fructose
Column:	ProntoGel Su Ca
Dimension:	300mm x 8mm
Mobil Phase:	H_2O
Flow:	0.5ml/min
Temperature:	80°C
Detection:	RI
Injection:	20µl sample



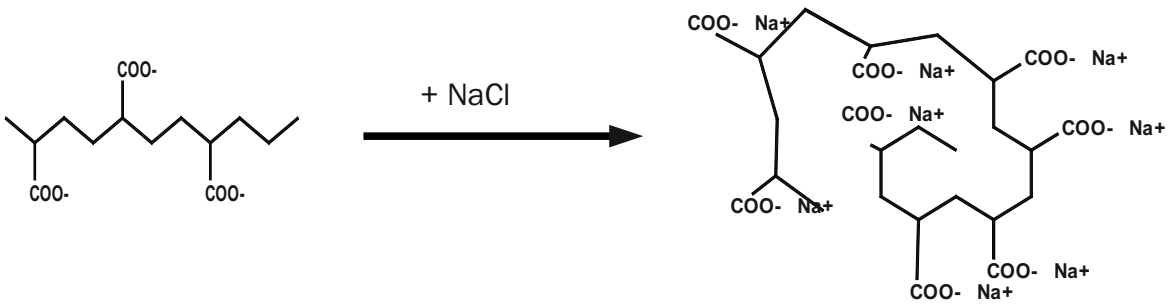
Analyte:	Honey
Column:	ProntoGel Su Na
Dimension:	300mm x 8mm
Mobil Phase:	H_2O
Flow:	0.5ml/min
Temperature:	80°C
Detection:	RI
Injection:	20µl sample

ProntoGel-Oligo-Su

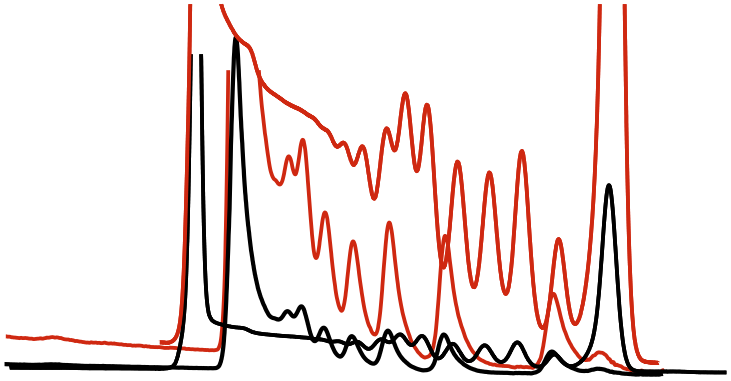


α–1 , 4

Description	Dimesnsion	
ProntoGel Oligo-Su Na	300mm x 8mm 250mm x 8mm 150mmx8mm 50mm x 8mm 30mm x 8mm	Oligosaccharide analysis in water
ProntoGel Oligo-Su Ag	300mm x 8mm 250mm x 8mm 150mmx8mm 50mm x 8mm 30mm x 8mm	Oligosaccharide analysis in water



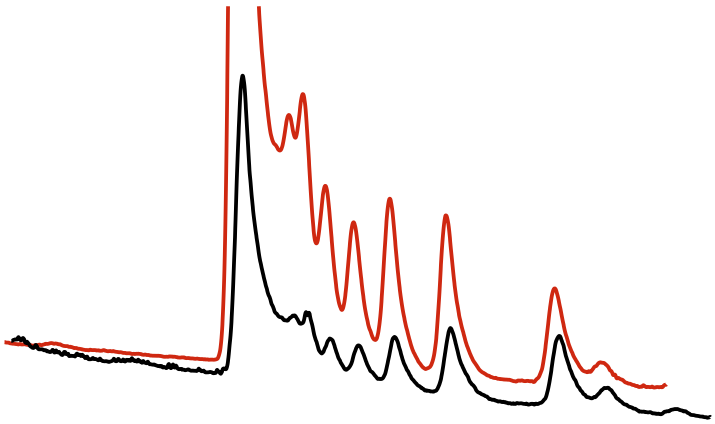
Oligosaccharide analysis in water



Analyte: **Maltodextrin 19**

Detailed view

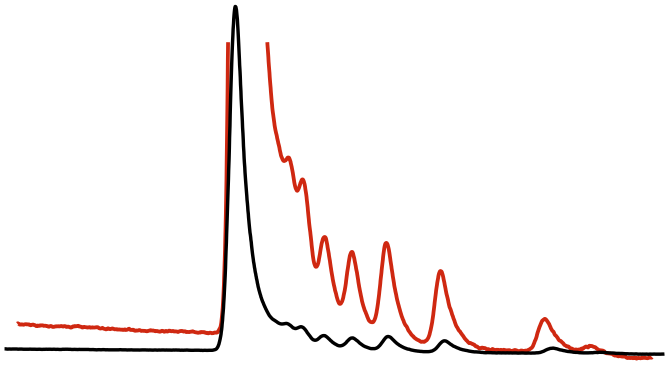
Column:	ProntoGel Oligo-Su Na
Dimension:	300mm x 8mm
Mobil Phase:	H_2O
Flow:	0.25ml/min
Temperature:	70°C
Detection:	RI
Injection:	20µl sample



Analyte: **Maltodextrin 12**

Detailed view

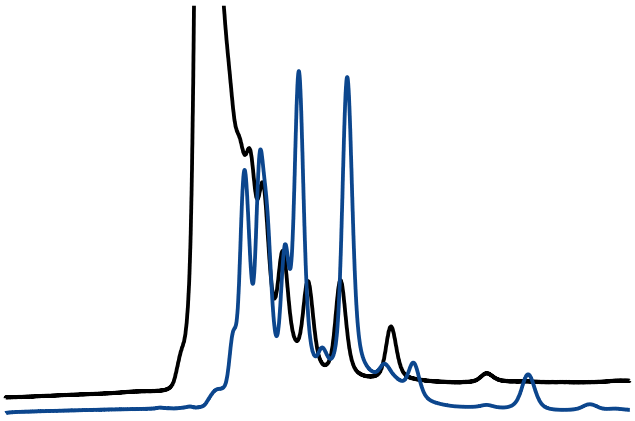
Column:	ProntoGel Oligo-Su Na
Dimension:	300mm x 8mm
Mobil Phase:	H_2O
Flow:	0.25ml/min
Temperature:	70°C
Detection:	RI
Injection:	20µl sample



Analyte: **Maltodextrin 6**

Detailed view

Column:	ProntoGel Oligo-Su Na
Dimension:	300mm x 8mm
Mobil Phase:	H_2O
Flow:	0.25ml/min
Temperature:	70°C
Detection:	RI
Injection:	20µl sample



Analyte: **Maltodextrin 12 (black)**

Inulin (blue)

Column:	ProntoGel Oligo-Su Na
Dimension:	300mm x 8mm
Mobil Phase:	H_2O
Flow:	0,25ml/min
Temperature:	70°C
Detection:	RI
Injection:	20µl sample

High resolutiom, easy detection with RI-Detector

ProntoGel-P

ProntoGel P columns for GPC analysis of organic molecules using (THF, toluene, chloroform).**
Oligomers and polymers including the new GPC media line from ProntoGel BPT* synthesis technology
for large range, high resolution separations with increased accuracy in calibration

- Spherical high porous styrene-divinyl benzene GPC-media
- Large molecular weight range: 100→10 000 000Da
- Optimized for 1ml/min flow rate when using 8mm ID-columns
- High pressure stability of 150-50 bar, depending on porosity
- High capacity from ProntoGel 8mm ID GPC columns
  (5% more capacity than 7.8mm columns, 14% more capacity than 7.5mm ID columns)
- Plus extra high pore volume from ProntoGel GPC synthesis technology for an extra increasing of peak capacity and resolution
- Proprietary ProntoGel GPC column packing procedure for accurate peak performance, low back Pressures and an extension of column livetime
- Long column live time for reduction of costs even at high through put screening applications
  High level of reproducibility
- High purity of ProntoGel GPC particles and columns for pure GPC mechanisms, low signal noise and reduction of “systempeaks” after GPC run

Area of applications:

epoxid resins, oligomers, isocyanates, PMMA / polymethylmethacrylate, polyethylmethacrylate, PS/polystyrene, vegetable oils /triglycerides/diglycerides,..., polybutadiene, polyisoprene, silicon / siliconoil / polydimethylsiloxane (in toluene), PEG / polyethylenglyol, polypropylenoxide, polyethylenglycol-polypropylenglycol-copolymer, PVC/polyvinylchloride, PU / polyurethane, celluloseacetate, diallylphthalate, dialkylphthalate, alkyd resin e.g...

GPC-examples (THF):

Amylose acetat, amylose propionat,
butyl rubber, cellulose diacetat, cellulosenitrat, polybutadiene, polycarbonate, polyisoprene, PMMA (polymethylmethacrylate), propylenglycol, polystyrene, polymethylstyrene, natural rubber, PVC (polyvinylchloride), polyvinylacetate, epoxid resins, polyisocyanate, polyols,
polyurethans, plant oils/triglycerids/diglycerids,....

GPC-examples

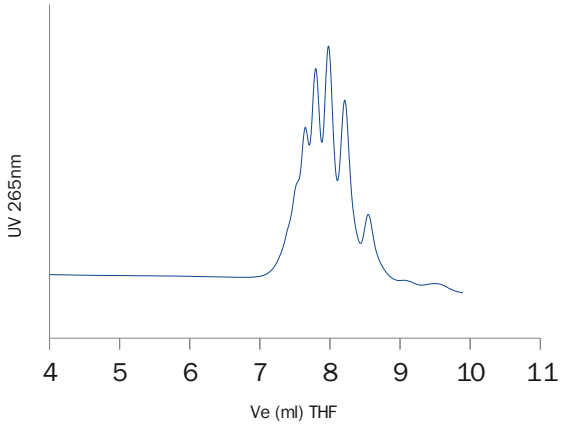
(toluene): Silicones, polydimethylsiloxan

ProntoGel-P molecular weight range and optimum range of molecular weights

ProntoGel-P-35	100Da-2 500Da
ProntoGel-P-100	100Da-10 000Da
ProntoGel-P-500	100Da-30 000Da
ProntoGel-P-1000	100Da-70 000Da
ProntoGel-P-1500	1.00Da-120 000Da
ProntoGel-P-10 000	100Da-400 000Da
ProntoGel-P-100 000	500Da-1 500 000Da
ProntoGel-P-1 000 000	10 000Da – 4 000 000Da
ProntoGel-P-10 000 000	100 000Da – 10 000 000Da

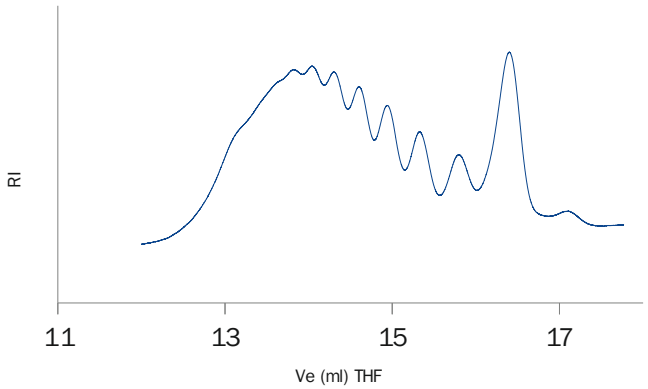
* Note:- (All above mentioned phases are available in 300x8, 250x8, 50x8, 30x8 dimension)

To cover a large range of molecular sizes GPC columns of suited porosities can be combined. This is the typical state of the art in many GPC laboratories. Also now it is useful if a special range of molecular sizes should be zoomed by GPC. But – in some single cases this also results in more or less obvious inhomogenities of calibration curves that itself makes accurate mathematics of calibration challenging. To increase accuracy and to simplify the calibration it uses the BPT* synthesis technology for significant lowering the inhomogenities phenomena and to improve the results.



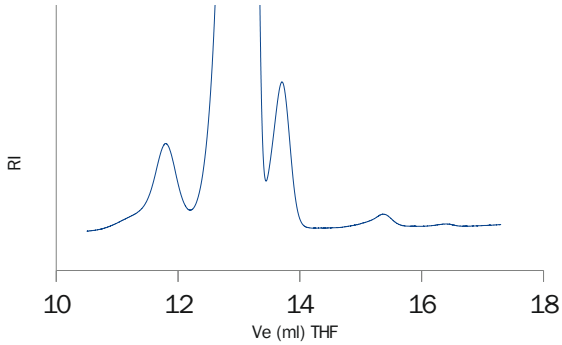
Analyte: **Polystryrene (PS) Mp = 578Da**

Column:	ProntoGel-P-35
Dimension:	300mm x 8mm
Mobil Phase:	THF
Flow:	1.0ml/min
Temperature:	20°C
Detection:	UV 265nm
Injection:	20µl sample



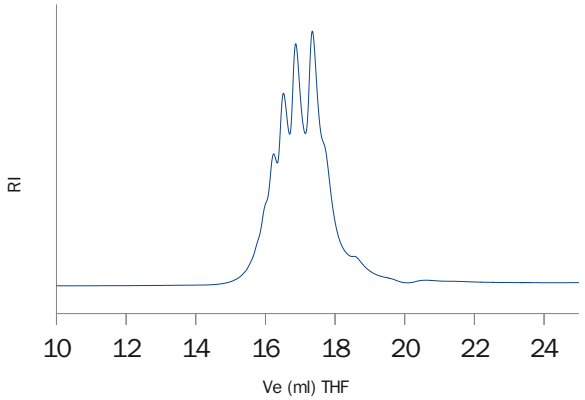
Analyte: **Polymethylmethacrylate (PMMA) oligomer**

Column:	ProntoGel-P-100
Dimension:	2 x 300mm x 8mm
Mobil Phase:	THF
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample



Analyte: **Grapeoil (Triglyceride GPC)**

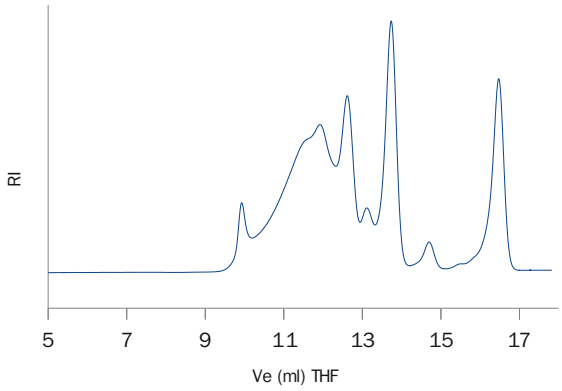
Column:	ProntoGel-P-100
Dimension:	2 x 300mm x 8mm
Mobil Phase:	THF
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample



Analyte: **Polyethylene glycol (PEG) Mp = 200Da**

Column:	ProntoGel-P-100
Dimension:	2 x 300mm x 8mm
Mobil Phase:	THF
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample

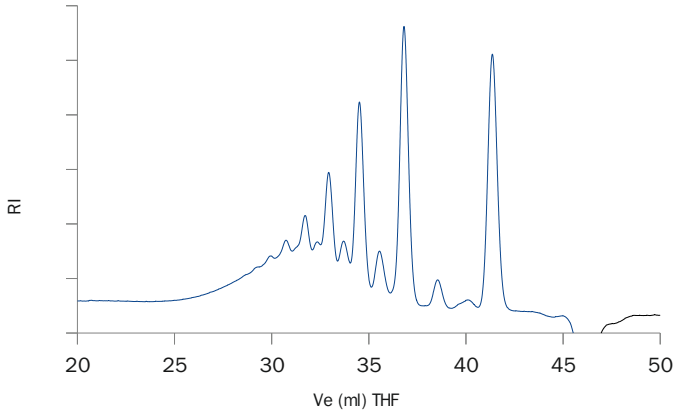
ProntoGel-P



Analyte: **Bisphenol-A-Epichlorhydrin resin (I)**

Column:	ProntoGel-P-100
Dimension:	2 x 300mm x 8mm
Mobil Phase:	THF
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample

Pore to small



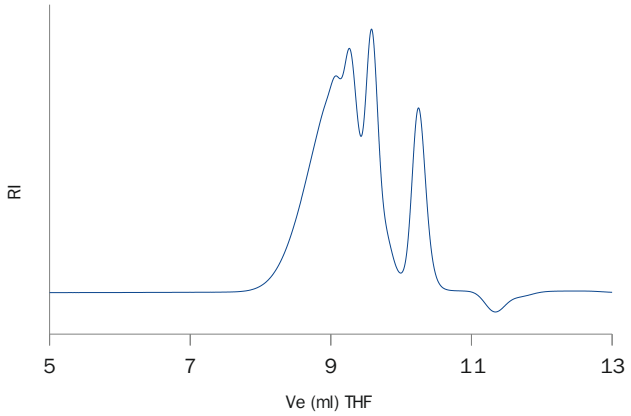
Pore perfect

High resolving GPC up to 10 000Da.

Analyte: **Bisphenol-A-Epichlorhydrin resin (I)**

Column:	ProntoGel-P-1500
Dimension:	2 x 300mm x 8mm
Mobil Phase:	THF
Flow:	0.5ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample

Good resolution, calibration range respective linearity in the range of 100Da-120 000Da, obtained from the special ProntoGel BPT-technology. No “surprising” porosity artefacts from mixing particles with pores of differnent size for covering the full range of molecular sizes. Great resolution even if 8µl RI measuring cell is used.

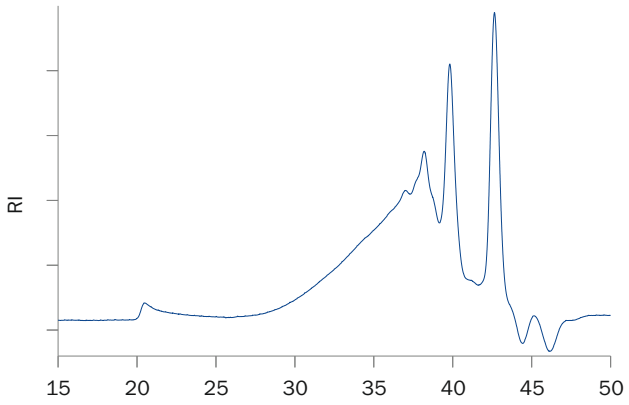


Pore to large

Analyte: **Bisphenol-A-Epichlorhydrin resin (I)**

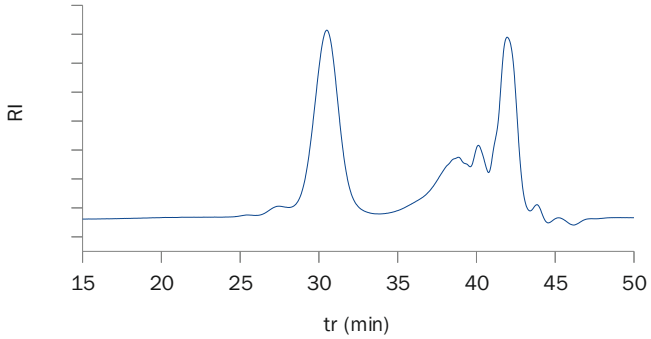
Column:	ProntoGel-P-100 000
Dimension:	300mm x 8mm
Mobil Phase:	THF
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample

Good linearity from monomer up to 1 500 000Da, ideal for porosity gap artefact reduced GPC screening of large ranges of molecular weights combined with maintaining of oligomer resolution



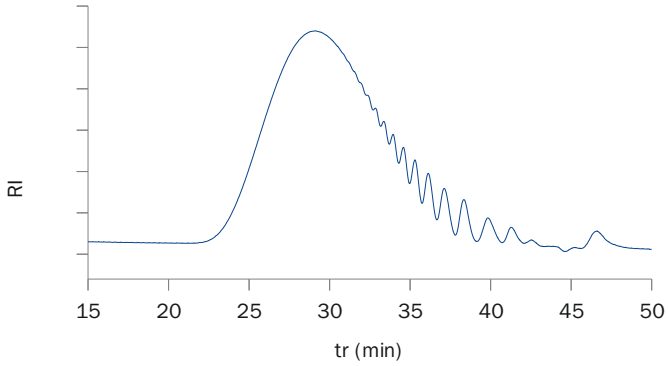
Analyte: **Polyol hard foam**

Column:	ProntoGel-P-1500
Dimension:	2 x 300mm x 8mm
Mobil Phase:	THF
Flow:	0.5ml/min
Temperature:	45°C
Detection:	RI
Injection:	20µl sample



Analyte: **Polyol soft foam**

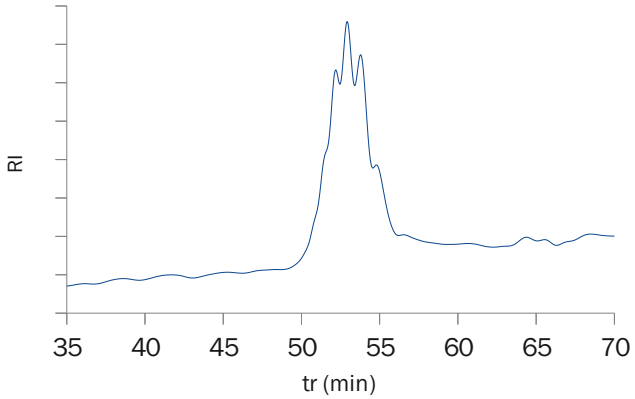
Column:	ProntoGel-P-1500
Dimension:	2 x 300mm x 8mm
Mobil Phase:	THF
Flow:	0.5ml/min
Temperature:	45°C
Detection:	RI
Injection:	20µl sample



Analyte: **Aliphatic polyester**

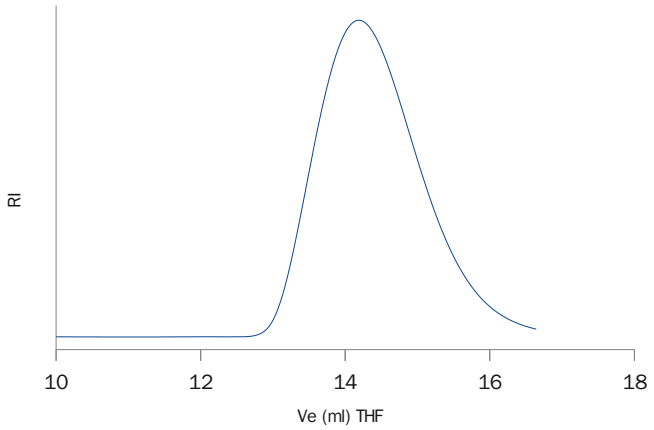
(adipate polyester)
including fingerprint

Column:	ProntoGel-P-1500
Dimension:	2 x 300mm x 8mm
Mobil Phase:	THF
Flow:	0.5ml/min
Temperature:	45°C
Detection:	RI
Injection:	20µl sample



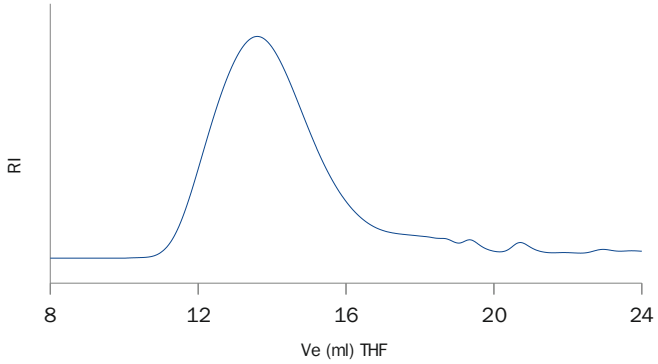
Analyte: **Polyether polyol**

Column:	ProntoGel-P-1500
Dimension:	2 x 300mm x 8mm 1x 50mmx 8mm
Mobil Phase:	THF
Flow:	0.5ml/min
Temperature:	45°C
Detection:	RI
Injection:	20µl sample



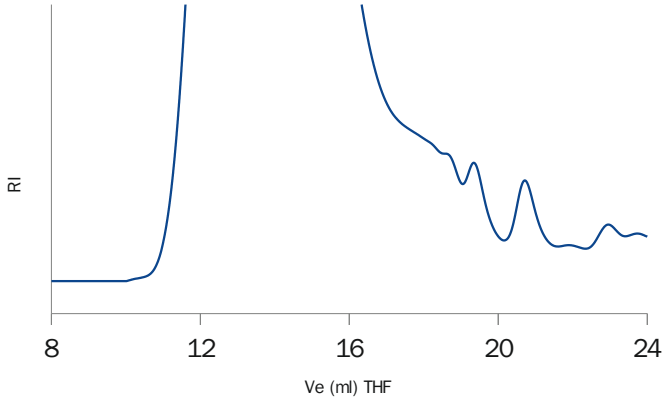
Analyte: **PVC**

Column:	ProntoGe-P-100 000
Dimension:	2 x 300mm x 8mm
Mobil Phase:	THF
Flow:	1.0ml/min
Temperature:	45°C
Detection:	RI
Injection:	20µl sample



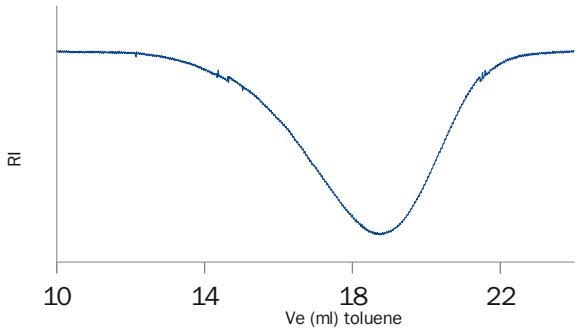
Analyte: **PMMA**

Column:	ProntoGel-P-100 000
Dimension:	2 x 300mm x 8mm
Mobil Phase:	THF
Flow:	1.0ml/min
Temperature:	45°C
Detection:	RI
Injection:	20µl sample



Analyte: **PMMA enlarged**

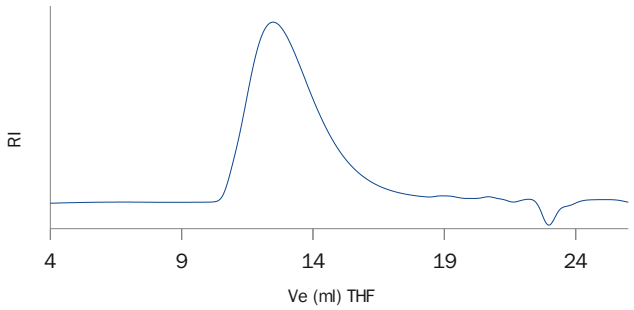
Column:	ProntoGel-P-100 000
Dimension:	2 x 300mm x 8mm
Mobil Phase:	THF
Flow:	1.0ml/min
Temperature:	45°C
Detection:	RI
Injection:	20µl sample



Analyte: **Silicone**

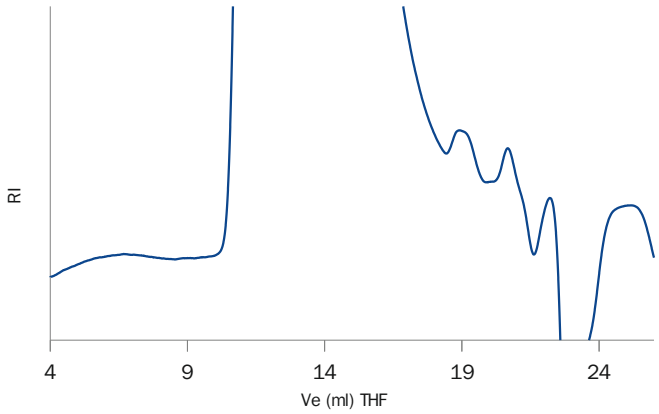
Column:	ProntoGel-P-100 000
Dimension:	2 x 300mm x 8mm
Mobil Phase:	Toluene
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample

Broad calibration range, no porosity artefacts observed



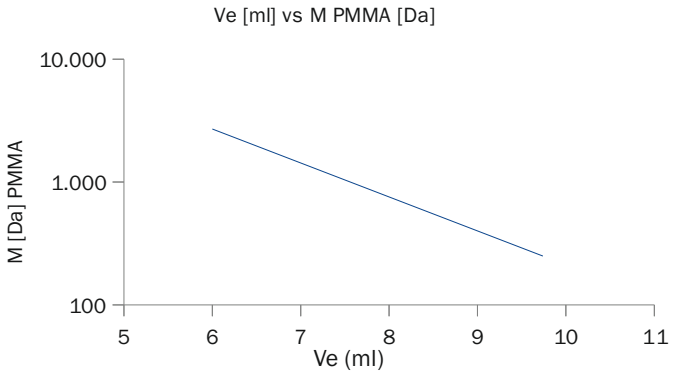
Analyte: **Polystyrene**

Column:	ProntoGel-P-100 000
Dimension:	2 x 300mm x 8mm
Mobil Phase:	THF
Flow:	1.0ml/min
Temperature:	45°C
Detection:	RI
Injection:	20µl sample



Analyte: **Polystyrene enlarged**

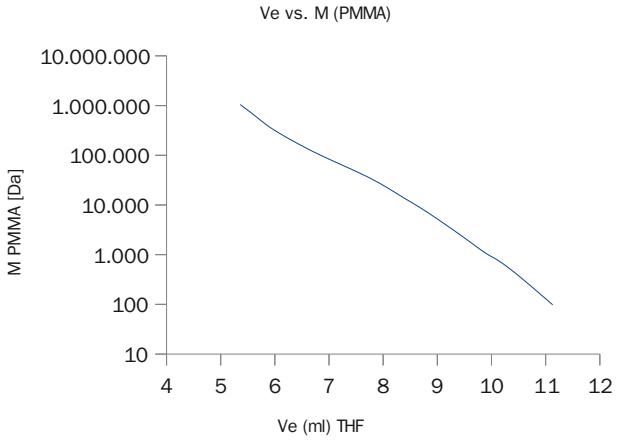
Column:	ProntoGel-P-100 000
Dimension:	2 x 300mm x 8mm
Mobil Phase:	THF
Flow:	1.0ml/min
Temperature:	45°C
Detection:	RI
Injection:	20µl sample



GPC calibration curve

Column:	ProntoGel-P-35 A
Dimension:	300mm x 8mm
Mobil Phase:	THF
Flow:	1.0ml/min
Temperature:	45°C
Detection:	RI
Injection:	20µl sample

GPC calibration cuve, large pore volume plus low exclusion limit for high oligomer resolution even withh low backpressure.



GPC calibration curve

Column:	ProntoGel-P-100 000
Dimension:	2 x 300mm x 8mm
Mobil Phase:	THF
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample

ProntoGel-P-100 000-BPT-Technology: Large calibration range respective good to calculate calibration curve from 100 000Da to 1 500 000Da, ideal technological step ahead if a combination of porosities results in “artificial shoulders" of calibration curves and/or in artificial shoulders of broad distributed molecular weight polymers. An easy to use tool to improve quality of results for your analytes.

ProntoGel-NAC

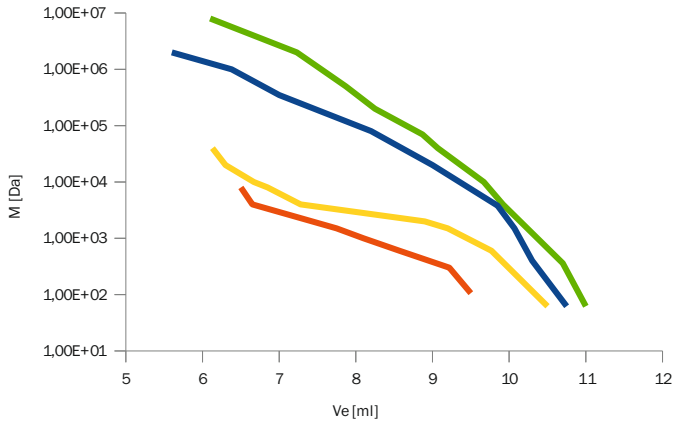
Special GPC / SEC columns for the aqueous analysis of neutral, anionic and additionally cationic polymers.

For GPC / SEC analysis of

- Polycations, polyammonium compounds
- polyamines (chitosans), polyethylenimines,
- PEGylated polyethyleneimines
- Polysaccharides
- Polyanions (heparins, pectins, ...)

Description	Separation Range
ProntoGel-100-NAC	100Da-2 500Da
ProntoGel-200-NAC	100Da-20 000Da
ProntoGel-350 NAC	2 500Da-1 000 000Da
ProntoGel-400-NAC	10 000Da-5 000 000Da

* Note:- (All above mentioned phases are available in 300x8, 250x8, 50x8, 30x8 dimension)

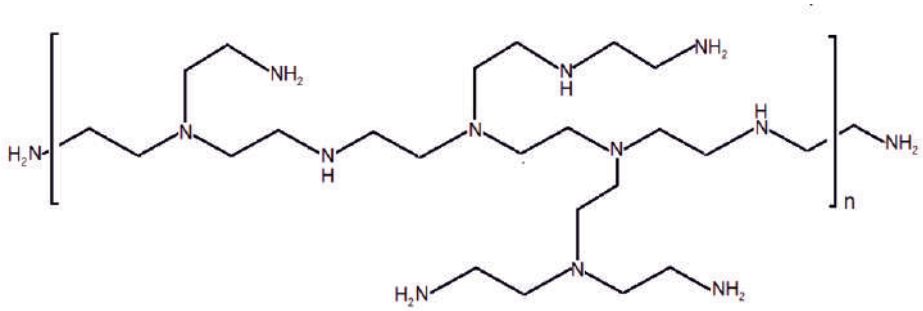


ProntoGel NAC Calibration curve
Porosity P-100, P-200, P-350 and P-400,
based on dextrans

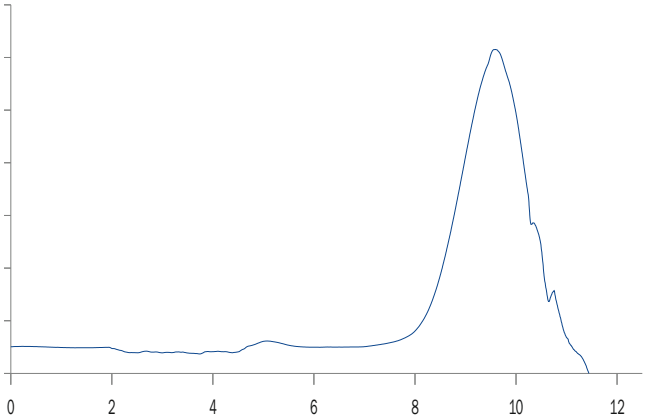
Analyte: **Dextrans**

Column: ProntoGel-100-NAC
ProntoGel-200-NAC
ProntoGel-350-NAC
ProntoGel-400-NAC

Dimension: e.a. 300mm x 8mm
Mobil Phase: H_2O,
Flow: 1.0ml/min
Temperature: 20°C
Detection: RI
Injection: 100µl sample



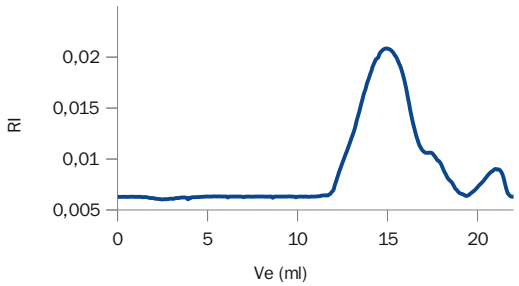
Polyethylenimine (PEI)



Analyte: Poly(2-vinylpyridine)
CAS [25014-15-7], Mw = 40 000Da

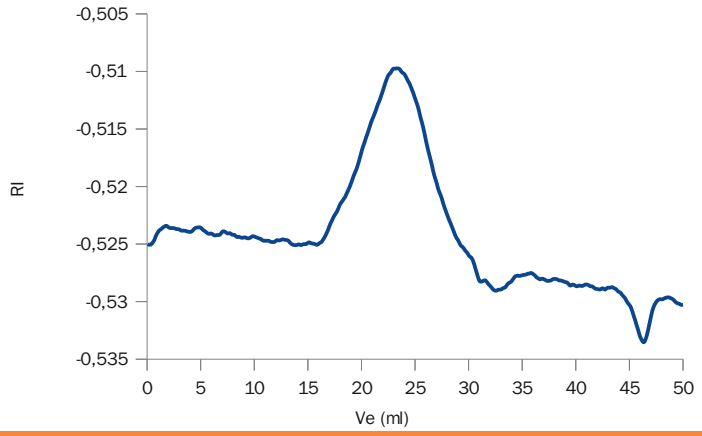
Column: ProntoGel-350-NAC

Dimension: 300mm x 8mm
Mobil Phase: 0.1M NaCl + 0.2% TFA in H_2O,
Flow: 1.0ml/min
Temperature: 20°C
Detection: RI
Injection: 20µl sample



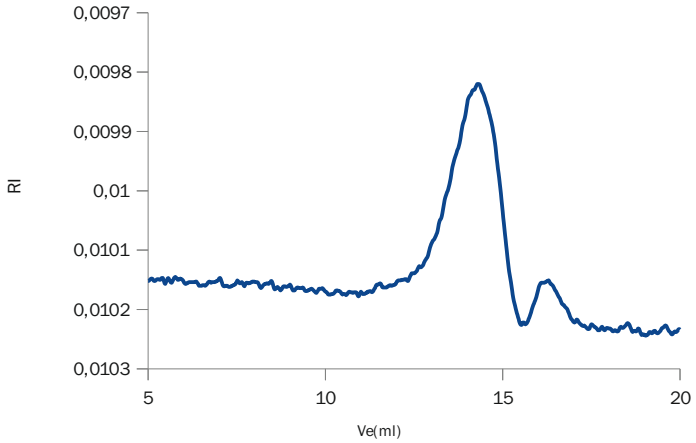
Analyte: **Poly DADMF**
Polydiallyldimethylammonium chloride, polyquaternium-6
Mw = 100-200 000Da, CAS [26062-79-3]

Column:	ProntoGel-100-NAC ProntoGel-350-NAC
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	$NaNO_3$ + 0.2% formic acid in H_2O
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	100µl sample



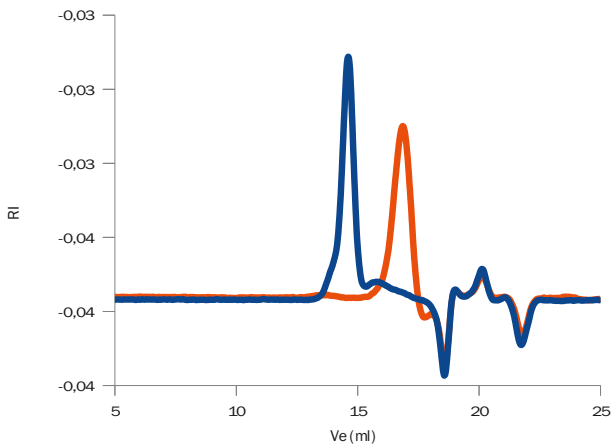
Analyte: **High molecular weight chitosan, poliglusam, polyglucosamine**
CAS [9012-76-4]

Column:	ProntoGel-400-NAC
Dimension:	3 x 300mm x 8mm
Mobil Phase:	$NaNO_3$ + 0.2% formic acid in H_2O
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	100µl sample



Analyte: **Chitosan sulfate**

Column:	ProntoGel-100-NAC ProntoGel-350-NAC
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	0.05MNa_2HPO_4 + 0.1M $NaNO_3$ in H_2O
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	100µl sample



Analyte: **PEI (polyethyleneimine)**
linear, PEI 150, CAS [9002-98-6]
Mn = 4 x 104, Mw = 4.2 x 104,
Mp = 3.9 x 104Da;

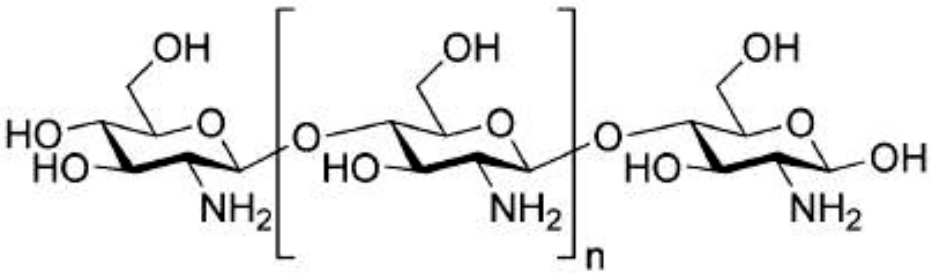
PEI (polyethyleneimine)
linear, PEI 25, CAS [9002-98-6]
Mn = 2.1 x 103, Mw = 2.9 x 103,
Mp = 2.15 x 103 Da

Column:	ProntoGel-100-NAC ProntoGel-350-NAC
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	0.1M NaCl + 0.2% TFA in H_2O
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	100µl sample

ProntoGel-NAC Plus

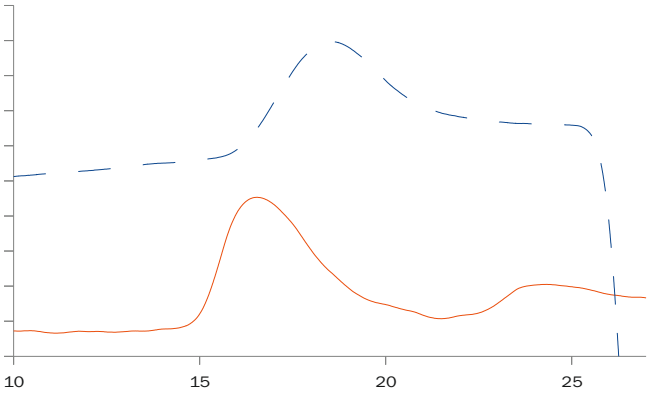
Description	Separation Range
ProntoGel-100-NAC Plus	100Da-2 500Da
ProntoGel-150-NAC Plus	100Da-5 000Da
ProntoGel-350-NAC Plus	2 500Da-1 000 000Da
ProntoGel-450-NAC Plus	300 000Da-50 000 000Da
ProntoGel-500-NAC Plus	10 000Da~50 000 000Da

* Note:- (All above mentioned phases are available in 300x8, 250x8, 50x8, 30x8 dimension)



ProntoGel-NAC Plus

Special GPC / SEC columns for the aqueous analysis of neutral, anionic and additionally cationic polymers - PLJ series with further increased hydrophilicity for extended application range - also in pure aqueous eluents (calibration with dextran / pullulan and new: additionally with PEO / PEG or p-2-vinylpyridine possible).

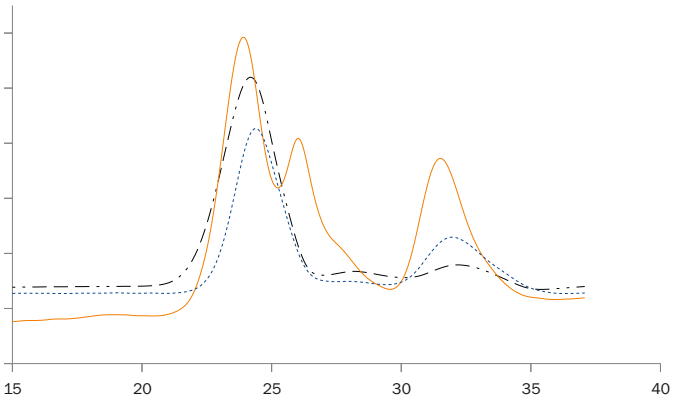


Analyte: **Polyquaternium-33**

very high molecular weight, copolymer of trimethylaminoethyl acrylate salt and acrylamide, CAS [69418-26-4] (red) vs. high molecular weight polyacrylamide, CAS [9003-05-8] 15Mio Da (blue, dashed)

Column:	ProntoGel-100-NAC Plus ProntoGel-350-NAC Plus ProntoGel-500-NAC Plus
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	0.1M NaCl + 0.2% TFA in H_2O,
Flow:	1.0ml/min
Temperature:	40°C
Detection:	RI
Injection:	100µl sample

GPC comparison 3 samples



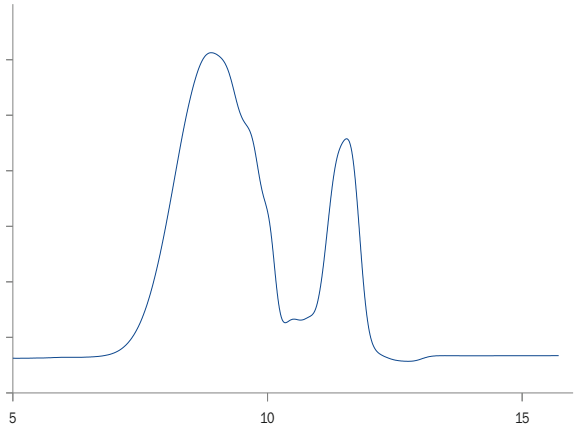
Analyte: **Poly DADMF**

Polydiallyldimethylammonium chloride, polyquaternium-6

Mw = 100Da-200 000Da,
CAS [26062-79-3]

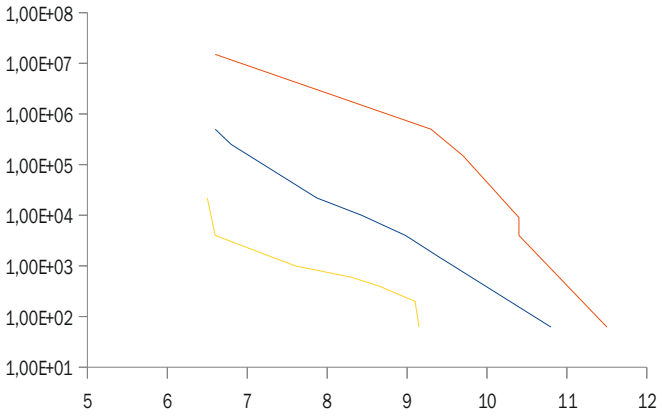
Column:	ProntoGel-100-NAC Plus ProntoGel-350-NAC Plus ProntoGel-500-NAC Plus
Dimension:	e.a. 300mm x 8mm
Mobil Phase:	0.1M NaCl + 0.2% TFA in H_2O,
Flow:	1.0ml/min
Temperature:	40°C
Detection:	RI
Injection:	100µl sample

ProntoGel-NAC Plus

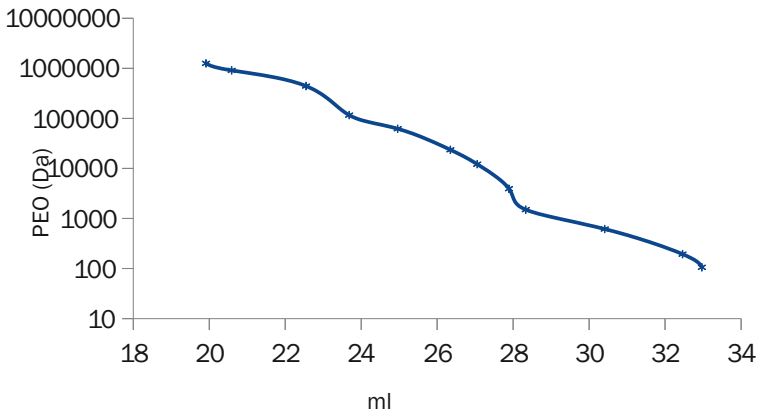


Analyte:	Poly(vinylimidazole), CAS [25232-42-2]
Column:	ProntoGel NAC Plus-350
Dimension:	300mm x 8mm
Mobil Phase:	0.1M NaCl + 0.2% TFA in H_2O,
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	20µl sample

ProntoGel NAC-PLJ Calibration curve
Porosity P-100-PLJ, P-350-PLJ and P-500-PLJ, based on dextrans



Analyte:	**PEO** **Dextran** **Polyacrylamide**
Column:	ProntoGel-100-NAC Plus ProntoGel-350-NAC Plus ProntoGel-500-NAC Plus
Dimension:	e.a.300mm x 8mm
Mobil Phase:	H_2O,
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI
Injection:	100µl sample



PEO/PEG Calibration

Column:	ProntoGel-100-NAC Plus ProntoGel-100-NAC Plus ProntoGel-350-NAC Plus ProntoGel-450-NAC Plus
Dimension:	1 x 50mm x 8mm 3 x 300mm x 8mm
Mobil Phase:	H_2O
Flow:	1.0ml/min
Temperature:	20°C
Detection:	RI

ProntoGel-HFIP

Benefits ProntoGel HFIP GPC columns series over standard columns

- Very large pore volume for high GPC resolution
- High GPC resolution for oligomers / condensates of 100Da 70 000Da
- High resolution GPC separation for the range 100Da-800 000Da
- Large areas with high linearity calibratable areas without porosity artifacts
- Low bleed
- Standard columns also used for GPC-LS and GPC viscosity

Description	Separation Range
ProntoGel-100-HFIP	100Da-2 500Da
ProntoGel-350-HFIP	up to 1 000 000 Da
ProntoGel-500-HFIP	10 000Da-~50 000 000Da

* Note:- (All above mentioned phases are available in 300x8, 250x8, 50x8, 30x8 dimension)

Special GPC / SEC columns for the GPC analysis of HFIP-soluble polymers.
For GPC / SEC analyzes from *)

Polyesters (polybutylene terephthalate / PBT / CAS 24968-12-5, polyethylene terephthalate /PET / CAS 25038-59-9, polylactide PLA / CAS 26100-51-6)

Polyamide 6 / PA6 / polycaprolactam / CAS 25038-54-4, polyamide 6-6 / PA6-6 / polyhexamethylene adipamide / CAS131-17-2, polyamide 6-10 / PA6-10 / poly(hexamethylene sebacamide) / PA 6-10, CAS 9011-52-3

Other (paraformaldehyde / polyoxymethylene / POM / polyacetal / CAS 30525-89-4 polyethylenimine / PEI / poly (iminoethylene / polyaziridine / CAS 9002-98-6)

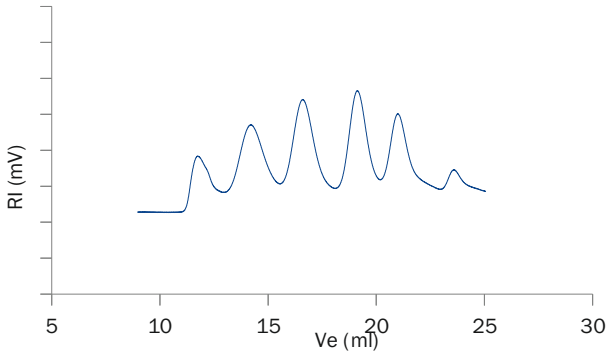
HFIP GPC / SEC calibration versus

- PMMA / polymethylmethacrylate / CAS 901-14-7

HFIP Accessories:

5mM CF3COOK or 0.075M CF3COONa are added to the eluent HFIP for the suppression of electrostatic interactions and for artefact-free GPC.

ProntoGel-HFIP

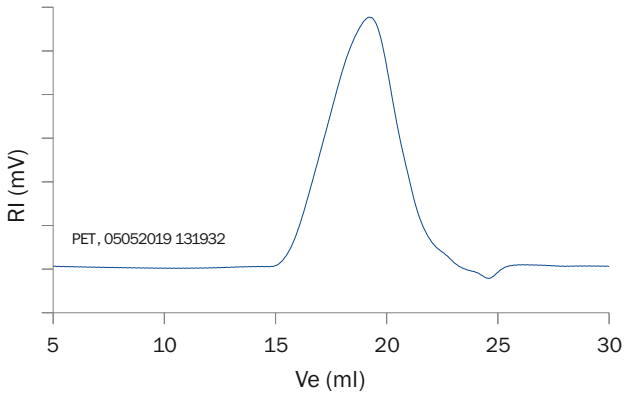


Analyte: **Polymethylmethacrylat**

(PMMA), CAS 9011-14-7, CAS131-17-2

M = 901.000, 96.760, 32.500, 3196, 540Da

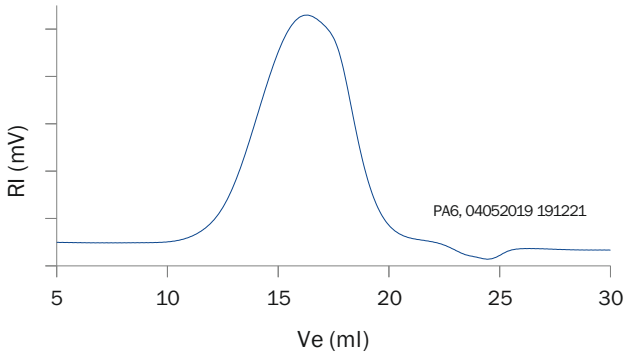
Column:	ProntoGel-350-HFIP
Dimension:	2 x 300mm x 8mm
Mobil Phase:	HFIP, 5mM CF_3 COONa
Flow:	0.5ml/min
Temperature:	40°C
Detection:	RI
Injection:	100µl sample



Analyte: **Polyethylenterephthalat**

(PET), CAS 25038-59-9

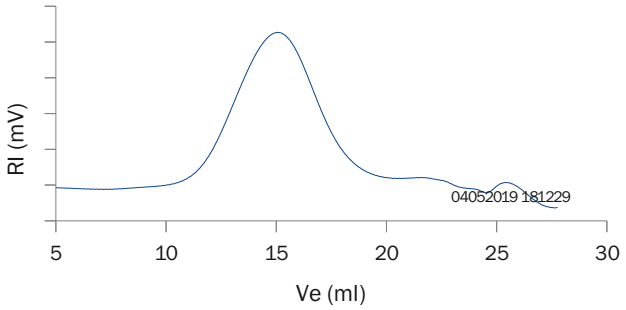
Column:	ProntoGel-350-HFIP
Dimension:	2 x 300mm x 8mm
Mobil Phase:	HFIP, 5mM CF_3 COONa
Flow:	0.5ml/min
Temperature:	40°C
Detection:	RI
Injection:	100µl sample



Analyte: **Polyethylenterephthalat**

(PET), CAS 25038-59-9

Column:	ProntoGel-350-HFIP
Dimension:	2 x 300mm x 8mm
Mobil Phase:	HFIP, 5mM Cf_3 COONa
Flow:	0.5ml/min
Temperature:	40°C
Detection:	RI
Injection:	100µl sample



Analyte: **Polyamide 6,6 (PA6-6),**

CAS131-17-2

Column:	ProntoGel-350-HFIP
Dimension:	2 x 300mm x 8mm
Mobil Phase:	HFIP, 5mM Cf_3 COONa
Flow:	0.5ml/min
Temperature:	40°C
Detection:	RI
Injection:	100µl sample; 1g/l

Good for many HFIP soluble polymers

ProntoGel-AC-AQ

ProntoGel AC-AQ-250, 300, 400 GPC-columns for GPC analysis of organic molecules using (80%aceton, 20%water)*.

Organosolv Lignin GPC analysis are now very easy and reliable to perform (Patent pending).

- Optimized for GPC analysis of organosolv lignins
- Easy and reliable to handle in aceton/water/traces formic acid // 80/20/1 // v/v/v
- Compatible with evaporative detection (ELSD, MS)*
- Preparative GPC fractionation of organosolv lignin without salt possible
- Recycling GPC respective peak recycling GPC with enormous separation efficiency possible for isolation of individual organosolv lignin substances in semipreparative scale
- Molecular weight calibration vs. PEO/PEG
- Spherical high porous polymeric GPC-media with no silanol activity for pure GPC
- Large molecular weight range: 100 – 1 000 000Da
- High pressure stability of 100bar
- High peak and separation capacity
- ProntoGel GPC columns – innovations and quality

Made in Germany

*) ask your ELSD respective your MS-detector supplyer for compatibility or your individual detector resistance versus aceton!

ProntoGel Ac-AQ GPC Series:

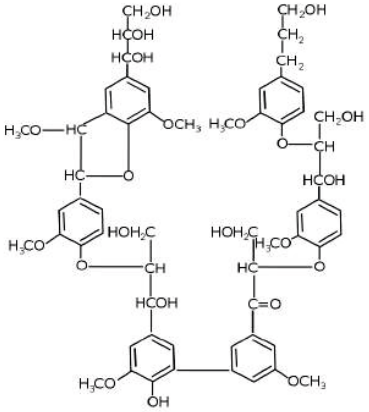
molecular weight range and optimum range of molecular weights

Pore size	Separation range	Optimum resolution range
250	100Da-70 000Da	100Da-10 000Da
350	100Da-1 000 000Da	1 000Da-400 000Da

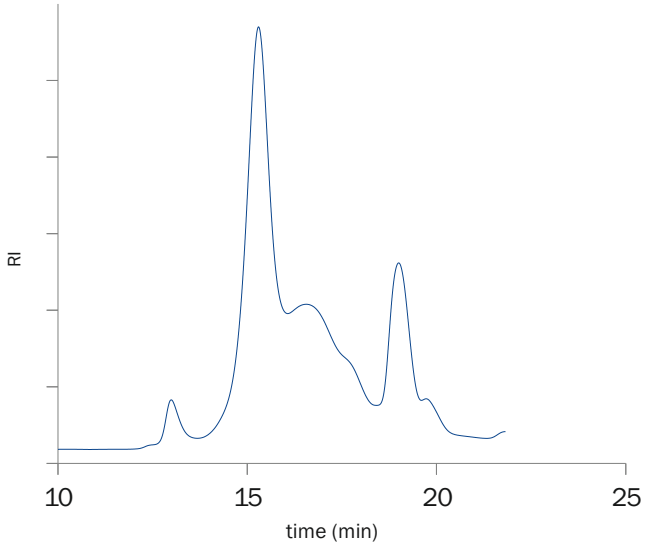
* Eluent 80/20 Aceton/waterv/v, molecular weight calibration vs. PEO/PEG

Description	Separation Range
GPC-Column ProntoGel-250-AC-AQ	**100Da–70 000Da**
GPC-Column ProntoGel-350-AC-AQ	**100Da–1 000 000Da**

* Note:- (All above mentioned phases are available in 300x8, 250x8, 50x8, 30x8 dimension)



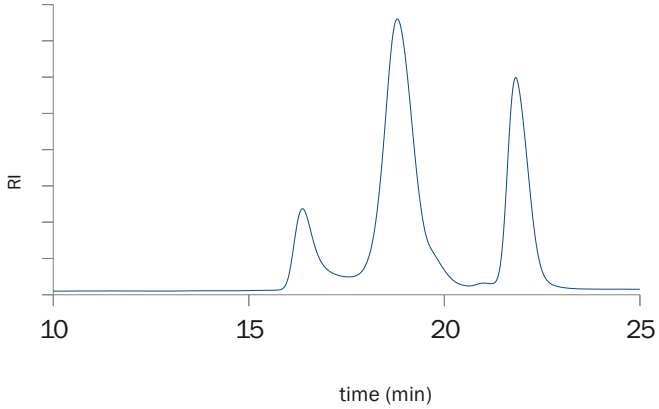
ProntoGel-AC-AQ



Analyte: **Organosolv Lignin GPC**
(range: 100Da-1 500 000Da)

Column: 2 x ProntoGel-250-AC-AQ

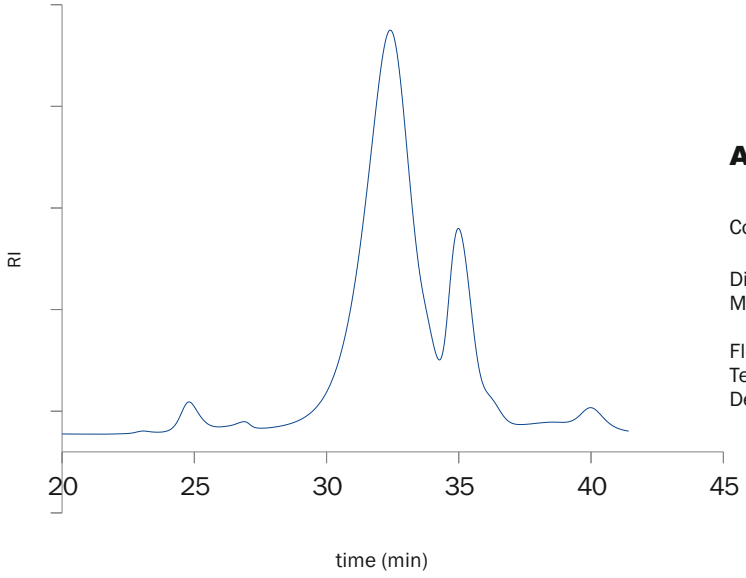
Dimension: ea. 300mm x 8mm
Mobil Phase: acetone/H_2O/formic acid 80/20/1 v/v/v
Flow: 1.0ml/min
Temperature: 25°C
Detection: RI



Analyte: **PEO/PEG GPC**
(8 000Da, 1 000Da, 106Da)

Column: 2 x ProntoGel-250-AC-AQ

Dimension: ea. 300mm x 8mm
Mobil Phase: acetone/H2O 80/20 v/v
Flow: 1.0ml/min
Temperature: 25°C
Detection: RI



Analyte: **Organosolv Lignin GPC**

Column: 3 x ProntoGel-350-AC-AQ

Dimension: ea. 300mm x 8mm
Mobil Phase: acetone/H2O/formic acid 80/20 /1 v/v/v
Flow: 1.0ml/min
Temperature: 25°C
Detection: RI

ProntoGel

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